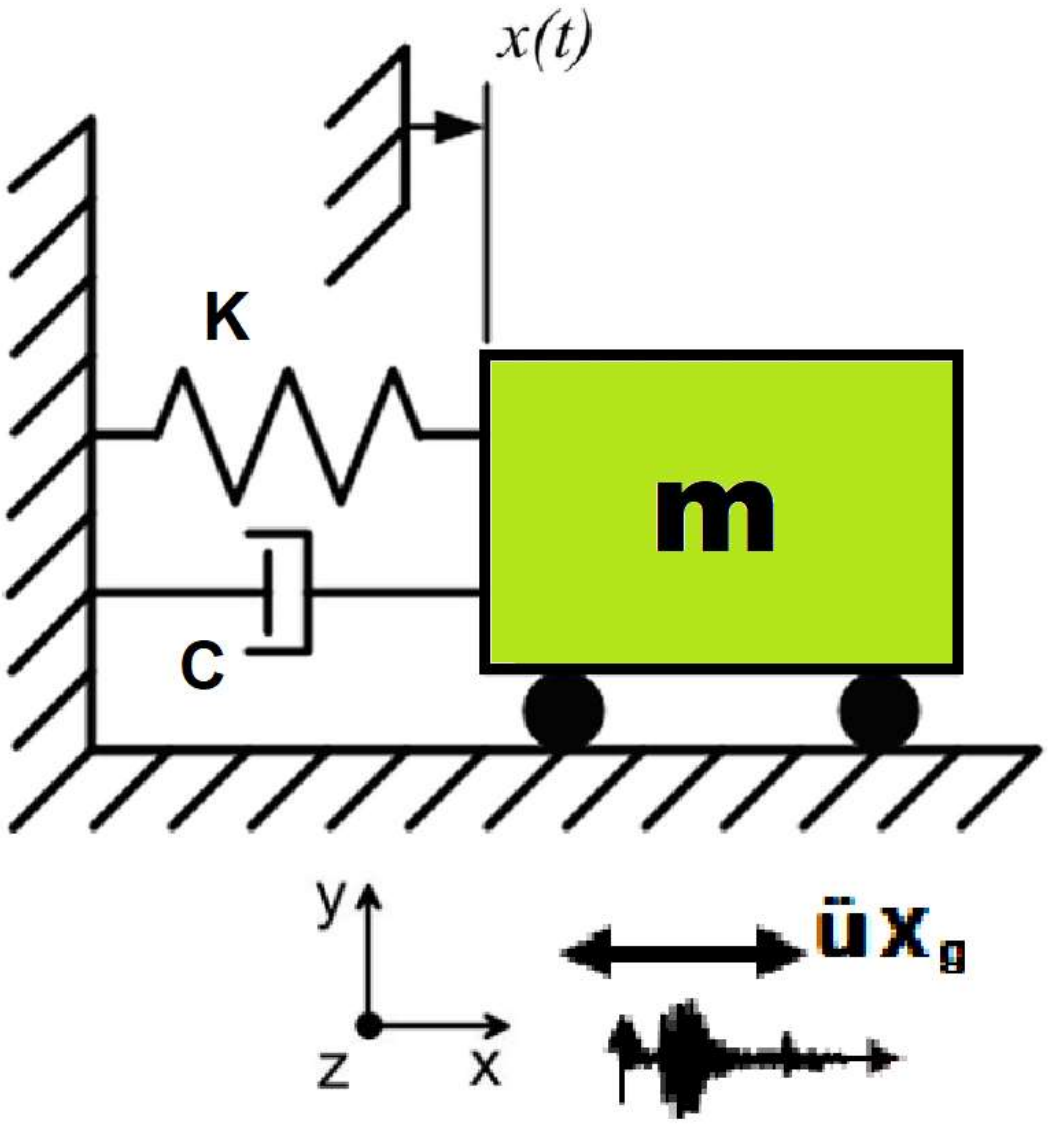
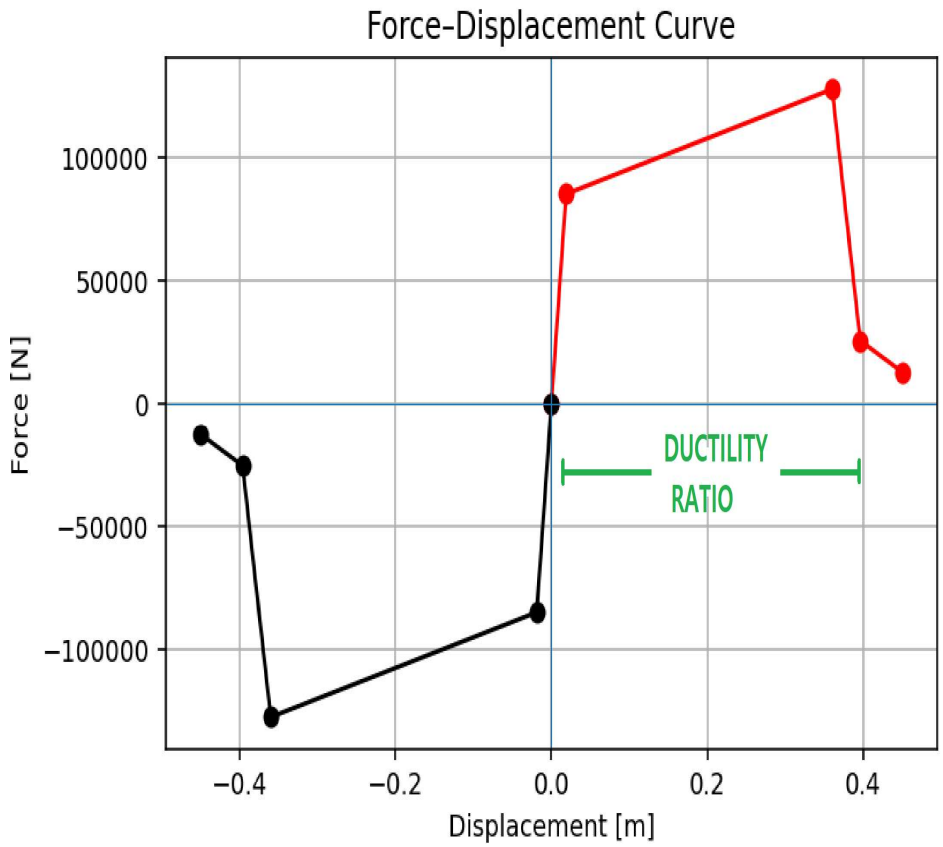
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

PARAMETRIC NONLINEAR SEISMIC DEMAND SPECTRA OF A SINGLE-DEGREE-OF-FREEDOM (SDOF) SYSTEMS WITH VARIABLE DUCTILITY CAPACITY IN OPENSEES

WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

This Python/OpenSeesPy script performs a parametric nonlinear time-history analysis of single-degree-of-freedom (SDOF) systems with constant yield strength and variable ductility capacity, first running a cyclic displacement protocol to establish a baseline dissipated energy capacity, then subjecting the same system to a selected ground motion while sweeping mass (and therefore period) to generate inelastic response spectra for displacement, velocity, acceleration, base reaction, ductility damage index, stiffness, damping ratio, energy dissipation capacity index, and equivalent viscous damping ratio; it computes damage and energy-based metrics, exports the peak response data to Excel, visualizes spectra with statistical summaries (mean, median, quartiles, standard deviation), performs fragility and machine-learning/correlation analyses, and plots time histories and hysteresis loops. Note, however, that despite the original header calling it a "constant ductility" response spectrum, the code does not iteratively scale the ground motion to enforce a target ductility demand; instead, it varies the ductility capacity of the backbone curve while keeping the seismic intensity and yield strength fixed, so it is more accurately described as a parametric nonlinear seismic demand spectrum study with variable ductility capacity.


x(t)
K
m
C
y
z
x
ü x g


Force-Displacement Curve
Force [N]
Displacement [m]
100000
50000
0
−50000
−100000
−0.4
−0.2
0.0
0.2
0.4
DUCTILITY
RATIO

$$Structural\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

$$\Delta_d = Lateral\ Displaement\ from\ Dynamic\ Analysis$$

$$\Delta_y = Lateral\ Yield\ Displaement\ from\ Pushover\ Analysis$$

$$\Delta_u = Lateral\ Ultimate\ Displaement\ from\ Pushover\ Analysis$$

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...CI_EVDR.+\SDOF_DUCT_RESPONSE_SPECTRUM_EDCI_EVDR.py

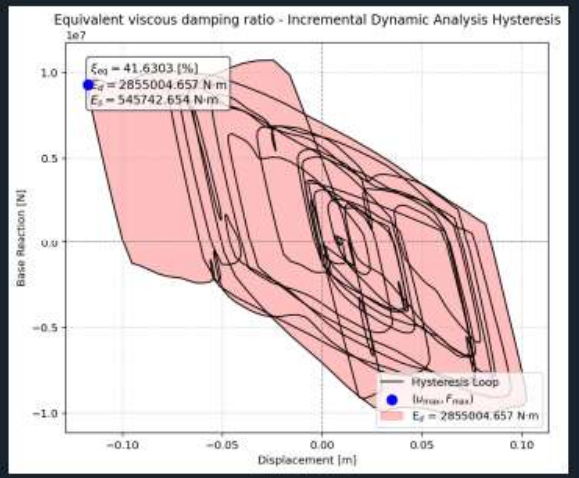
SDOF_DUCT_RESPONSE_SPECTRUM_EDCI_EVDR.py | EQULIVALENT_VISCOUS_DAMPING_RATIO_FUN.py

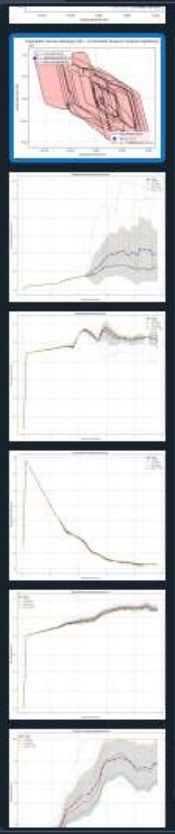
```
#########################################################################################################
#                    >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
#     NONLINEAR DYNAMIC ANALYSIS OF A SINGLE-DEGREE-OF-FREEDOM (MDOF) SOIL STRUCTURE SYSTEM UTILIZING
#          GROUND MOTIONS IN OPENSEES AND EVALUATION OF CONSTANT DUCTILITY SEISMIC RESPONSE SPECTRUM
#---------------------------------------------------------------------------------------------------------
#                                    INELASTIC RESPONSE SPECTRUM ANALYSIS
#---------------------------------------------------------------------------------------------------------
#                EQUIVALENT VISCOUS DAMPING RATIO: xi_eq = 100 * E_d / (4 * pi * E_s)
#---------------------------------------------------------------------------------------------------------
#          ENERGY DISSIPATION CAPACITY INDEX = 100 * E_d(earthquake) / E_d(cyclic displacement)
#---------------------------------------------------------------------------------------------------------
# This program performs  Dynamic Analysis on a Single-Degree-of-Freedom (SDOF) system
# subjected to 1 seismic ground motions. The analysis evaluates the structural response under varying
# levels of seismic intensity.
# The framework is designed to support researchers and engineers in assessing the probabilistic seismic
# performance of structures, with a focus on understanding the impact of uncertainty on structural
# response and design.
#---------------------------------------------------------------------------------------------------------
# Key Features:
# - Simulation of SDOF system using OpenSees.
# - Scaling of ground motions for Nonlinear Dynamic Analysis.
# - Probabilistic fragility assessment based on predefined damage states.
# - Visualization of structural response and fragility curves.
# - Export of results for further analysis.
#---------------------------------------------------------------------------------------------------------
#                         THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
#                                    EMAIL: salar.d.ghashghaei@gmail.com
#########################################################################################################

"""
This code implements a comprehensive nonlinear dynamic analysis framework for
performance-based earthquake engineering assessment of single-degree-of-freedom
(SDOF) systems. The methodology combines traditional nonlinear time-history
analysis with modern probabilistic and machine learning techniques for advanced
```

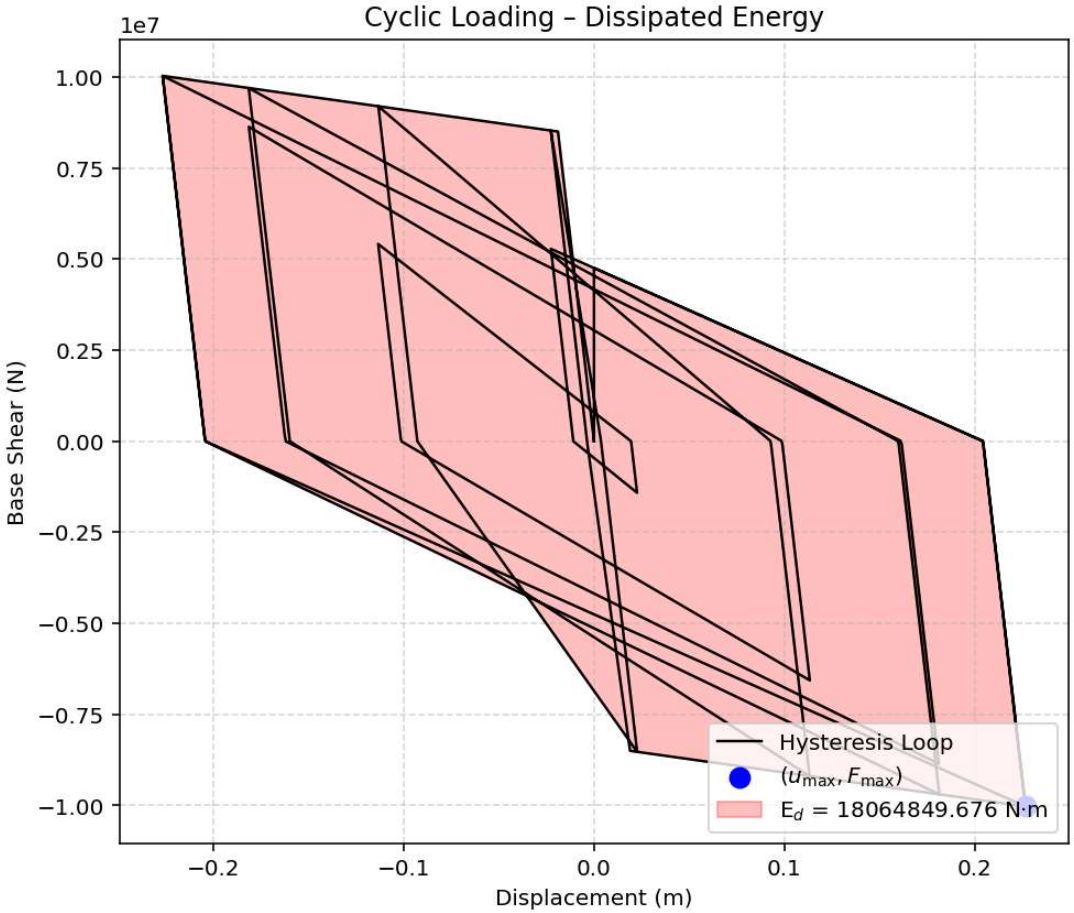
..DEX\11_SDOF_8_ANA\SDOF_DUCT_RESPONSE_SPECTRUM_EDCI_EVDR+

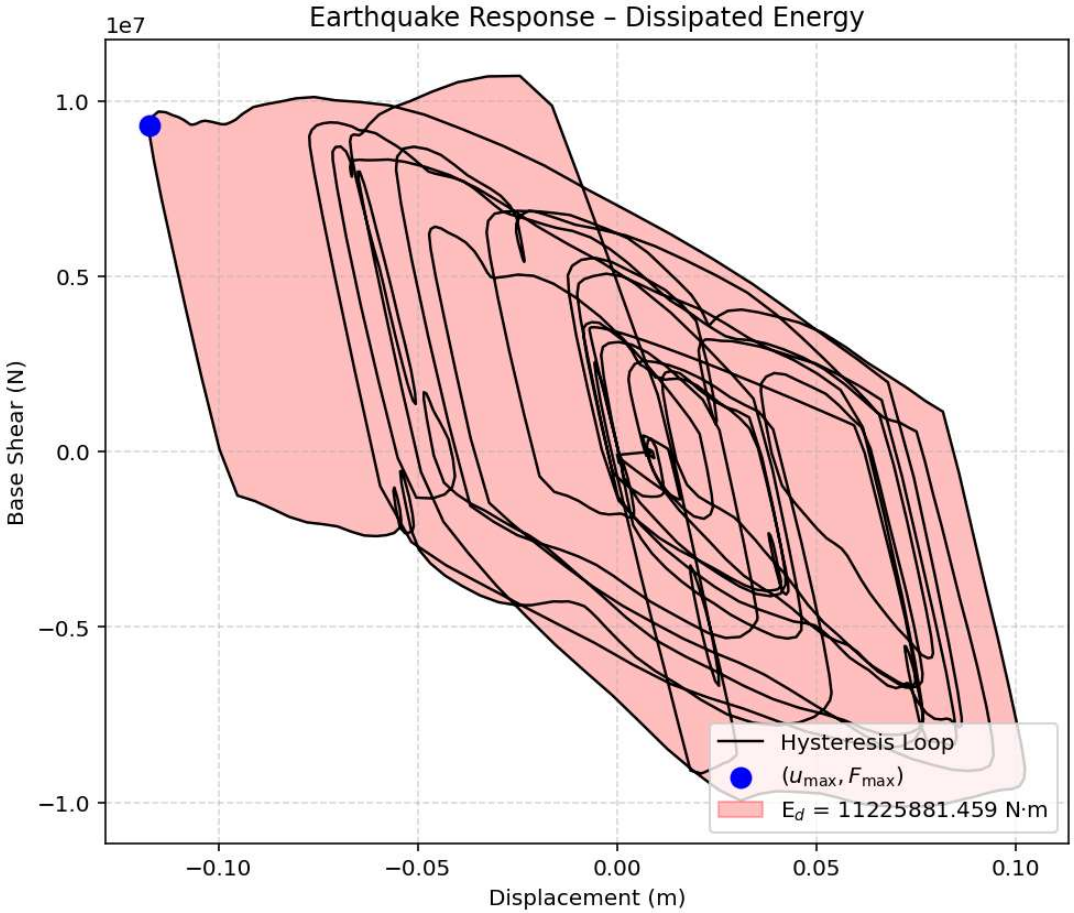
35 %

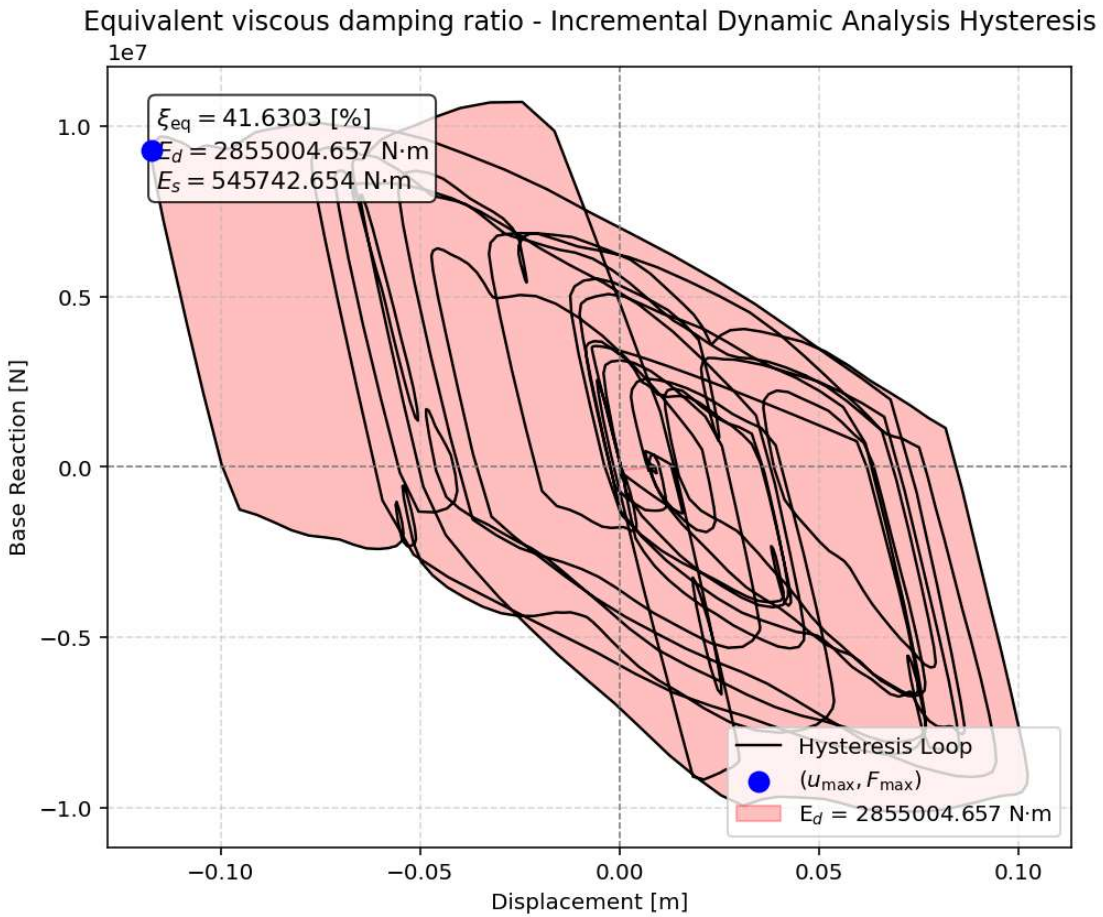


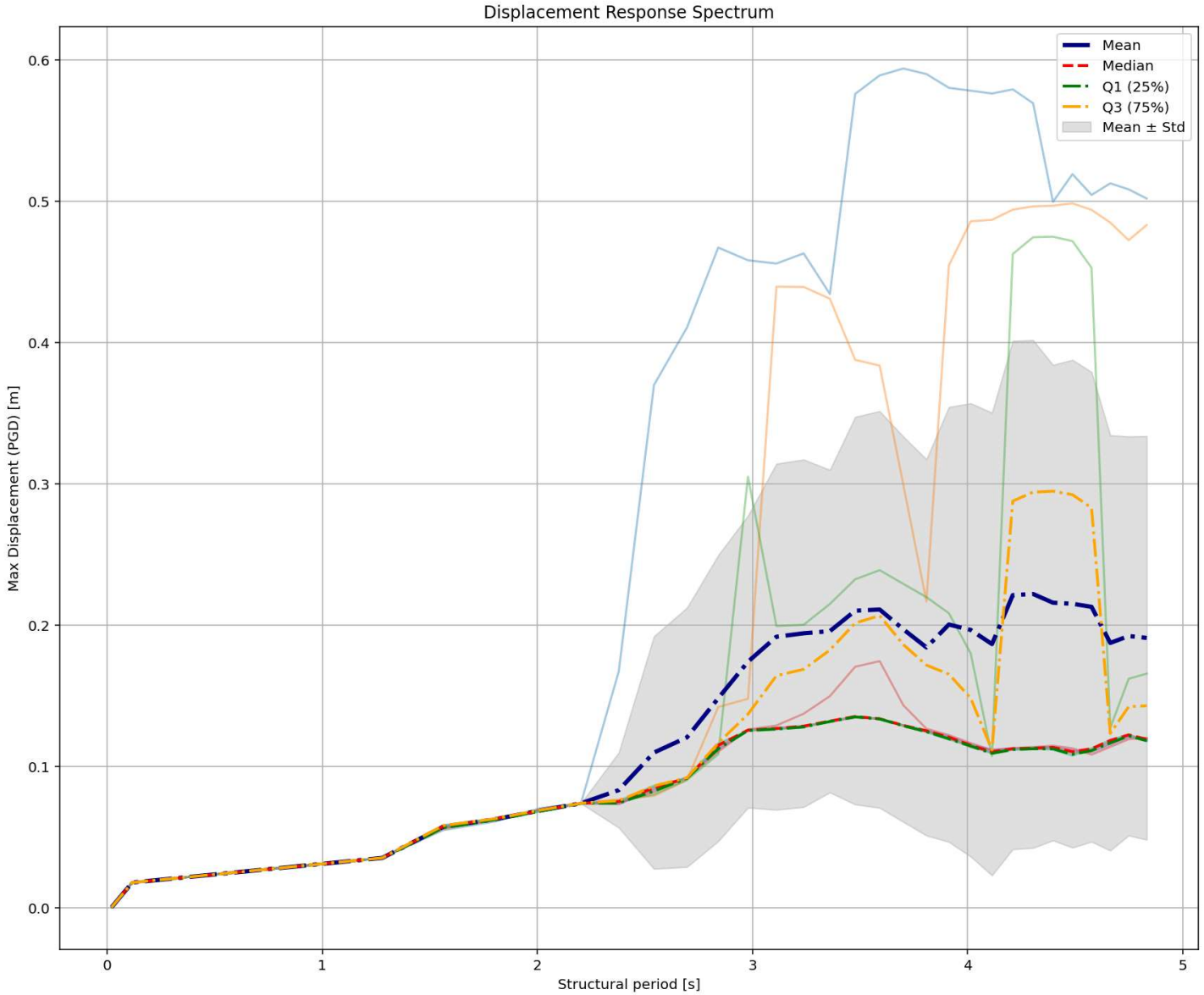


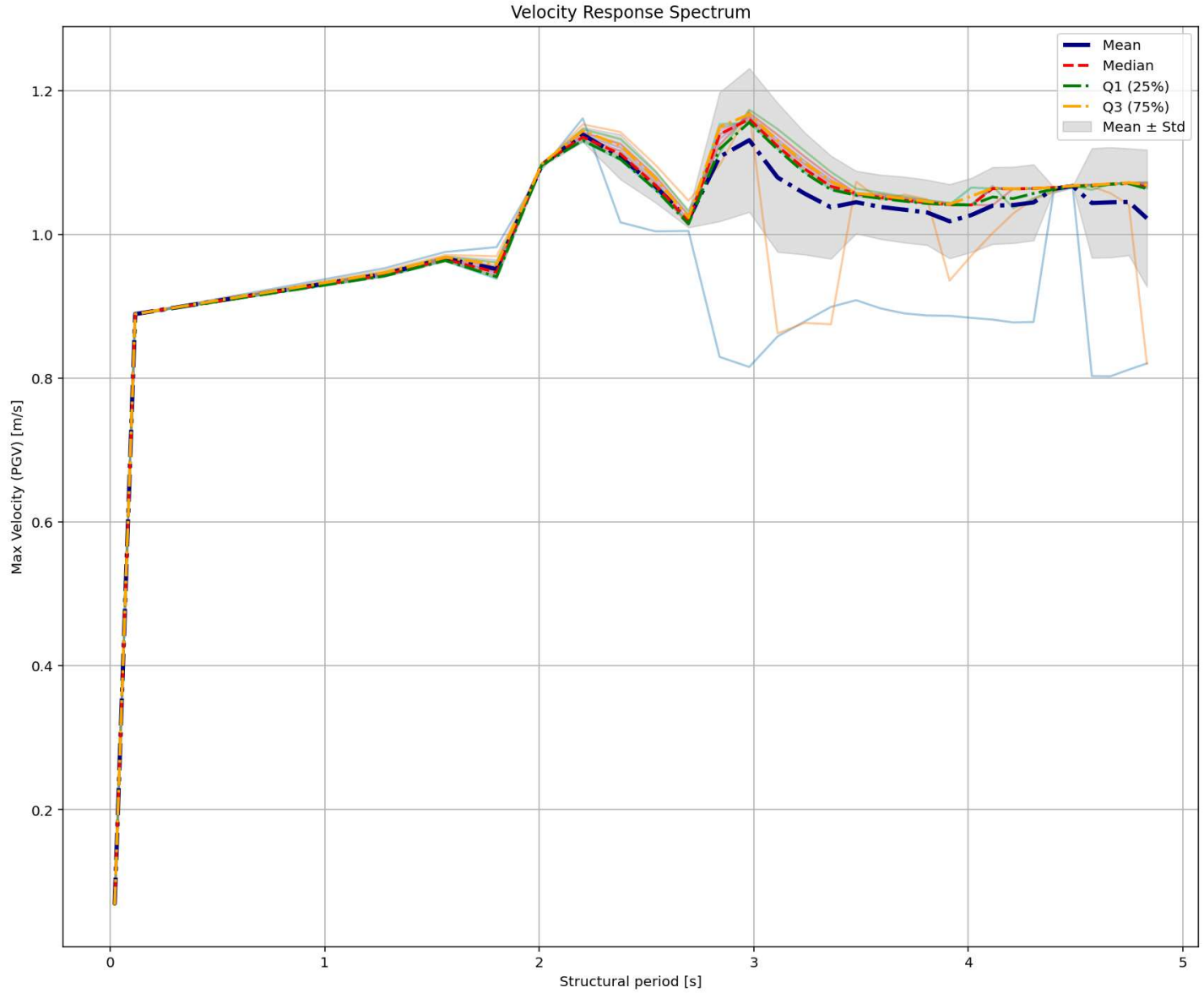
IPython Console | Plots | Variable Explorer | Help | Debugger | History | Files

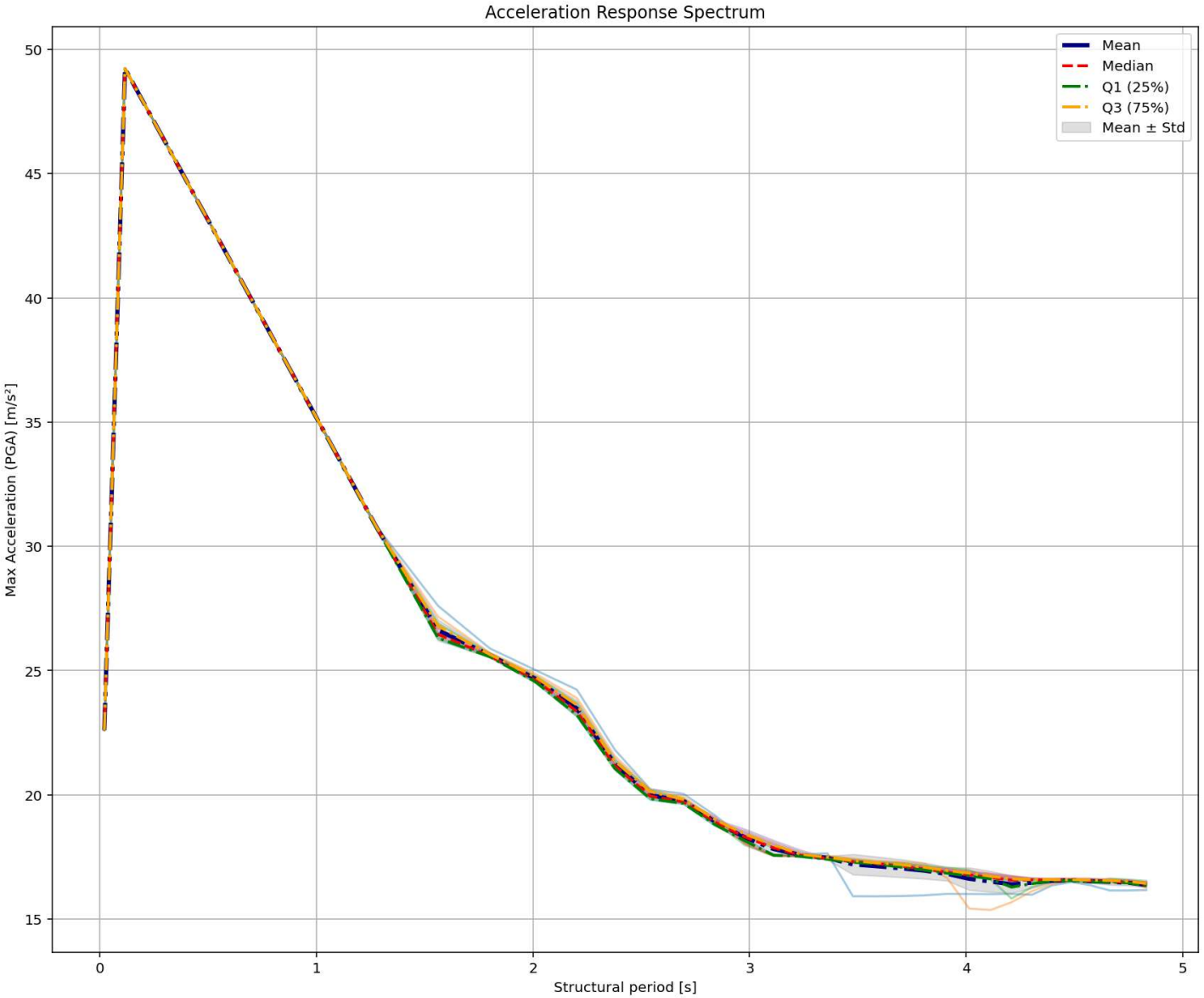

Cyclic Loading – Dissipated Energy
1e7
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Base Shear (N)
−0.2
−0.1
0.0
0.1
0.2
Displacement (m)
Hysteresis Loop
(u_{max}, F_{max})
E_d = 18064849.676 N·m

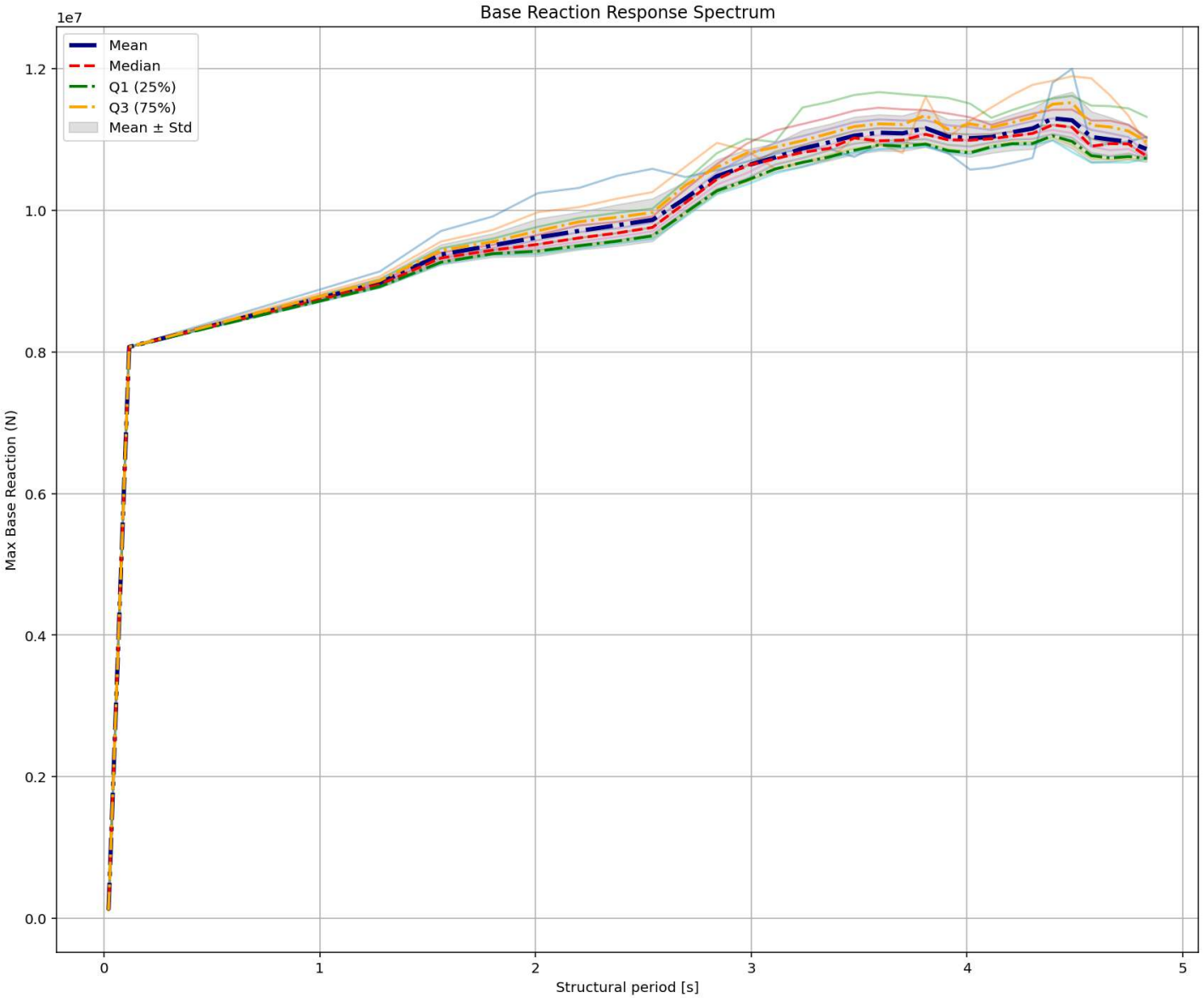

Earthquake Response – Dissipated Energy
1e7
1.0
0.5
0.0
−0.5
−1.0
Base Shear (N)
−0.10
−0.05
0.00
0.05
0.10
Displacement (m)
Hysteresis Loop
(u_{max}, F_{max})
E_d = 11225881.459 N·m

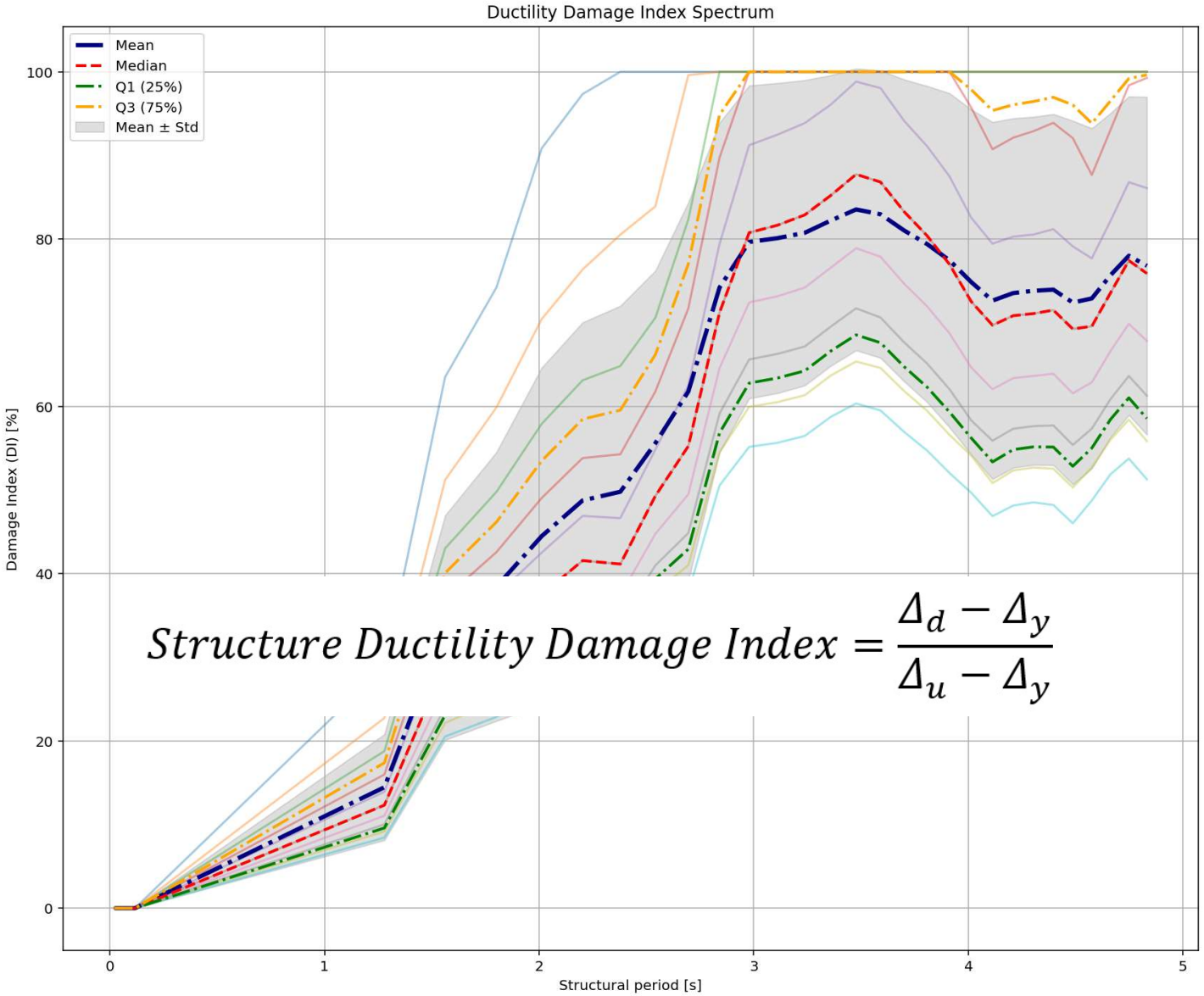

Equivalent viscous damping ratio - Incremental Dynamic Analysis Hysteresis
1e7
$\xi_{eq} = 41.6303$ [%]
$E_d = 2855004.657$ N·m
$E_s = 545742.654$ N·m
Base Reaction [N]
Displacement [m]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 2855004.657 N·m


Displacement Response Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Max Displacement (PGD) [m]
Structural period [s]
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0
1
2
3
4
5

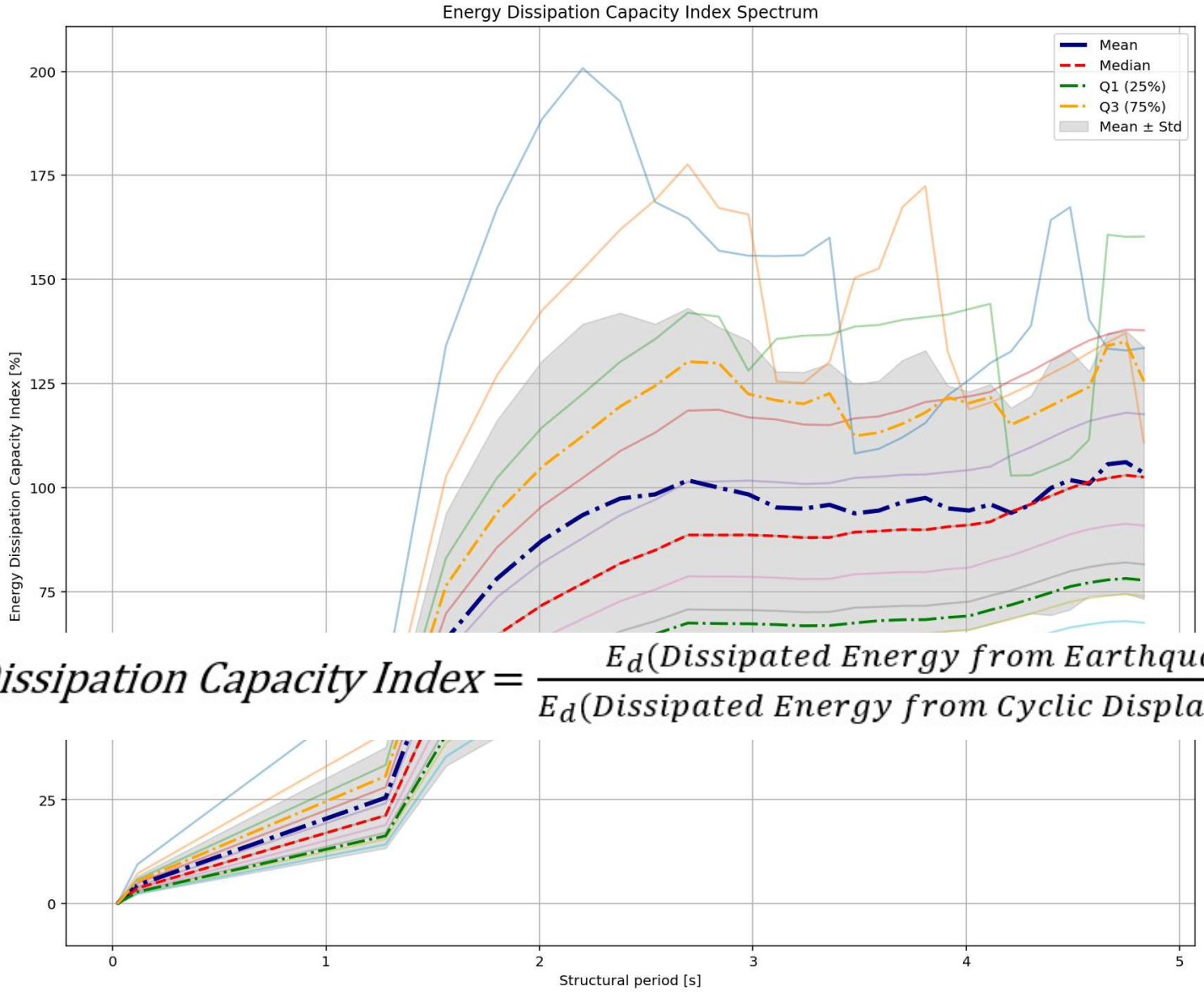

Velocity Response Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Max Velocity (PGV) [m/s]
Structural period [s]
0.2
0.4
0.6
0.8
1.0
1.2
0
1
2
3
4
5


Acceleration Response Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Max Acceleration (PGA) [m/s²]
Structural period [s]
50
45
40
35
30
25
20
15
0
1
2
3
4
5

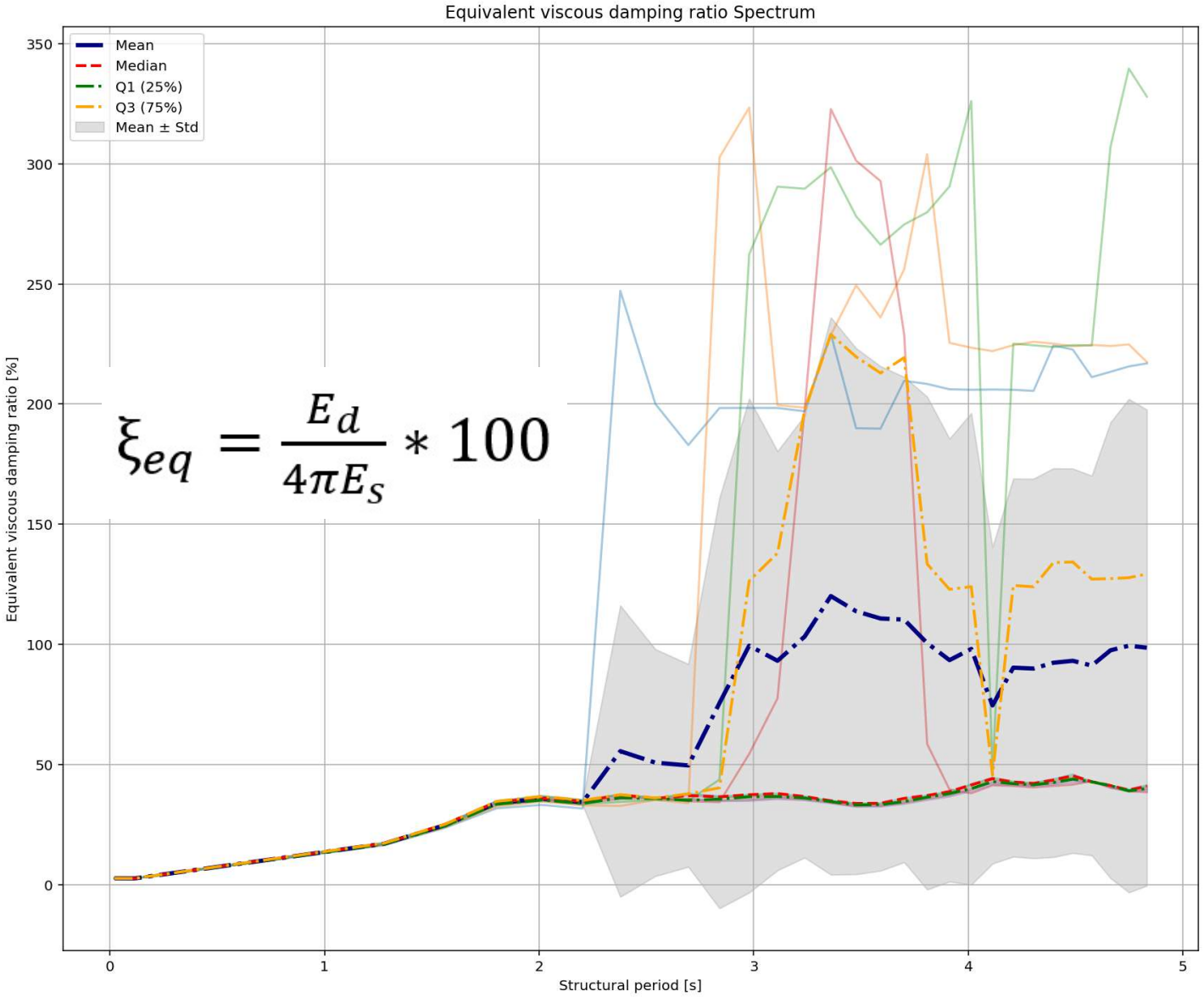

Base Reaction Response Spectrum
1e7
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
Max Base Reaction (N)
Structural period [s]
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0
1
2
3
4
5

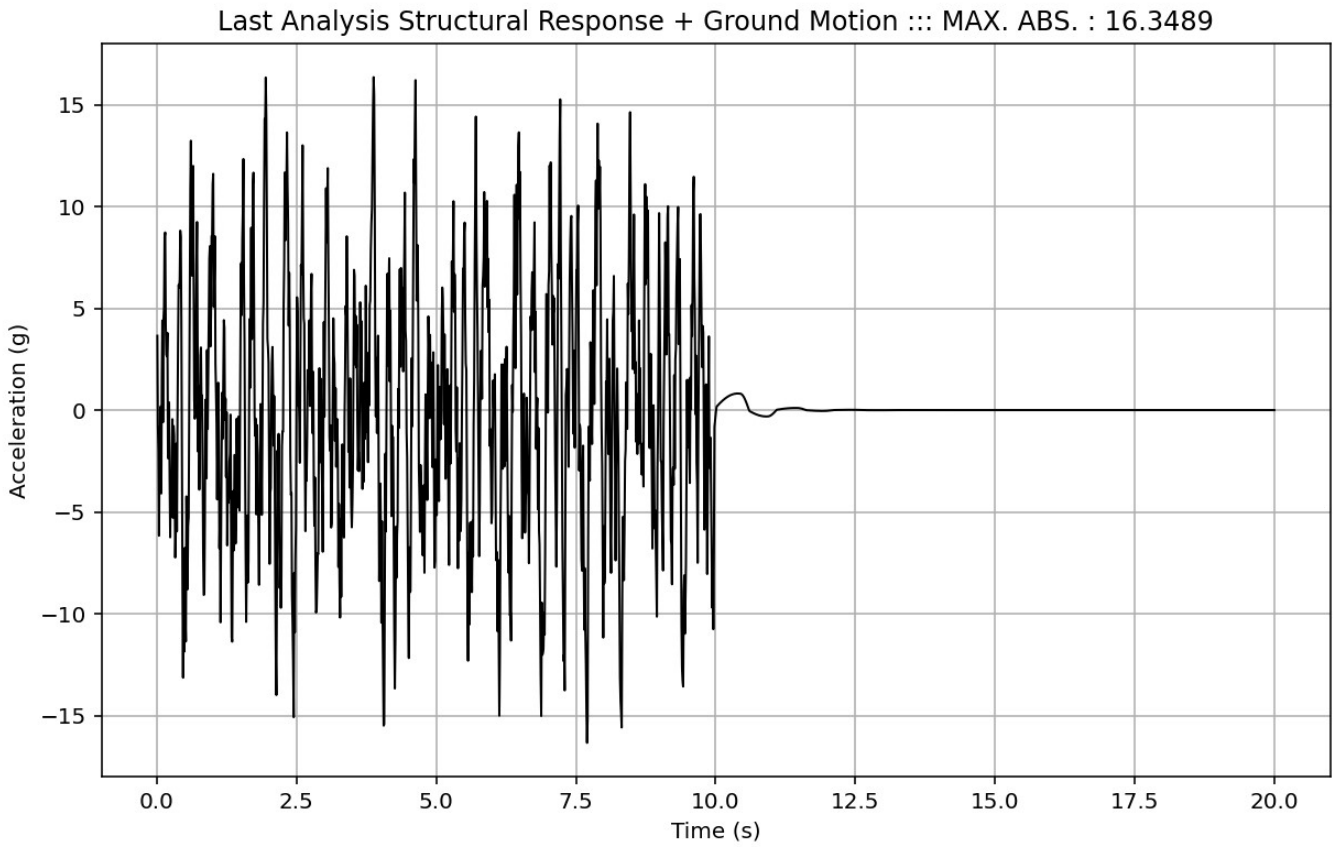


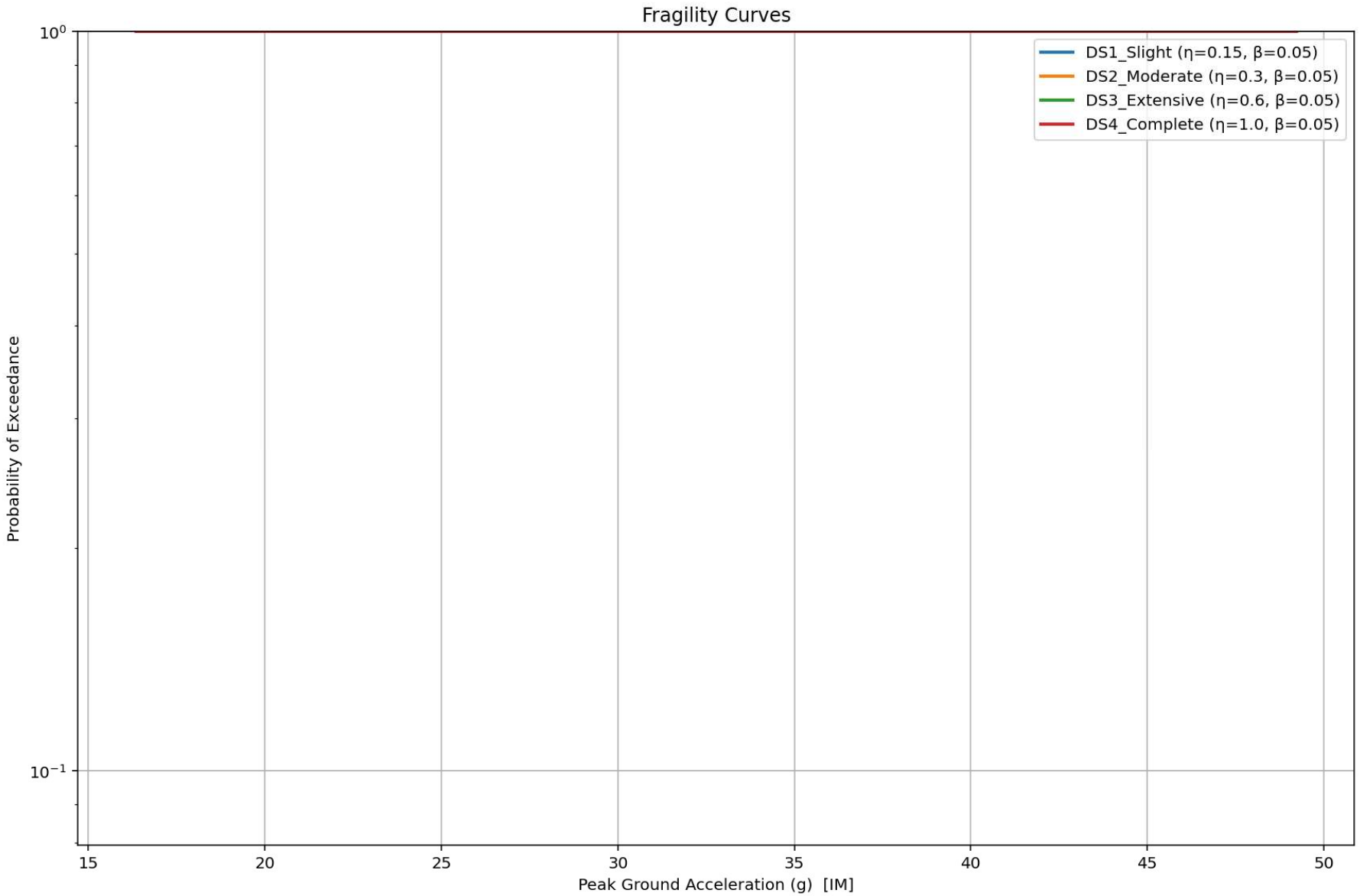
$$Structure\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

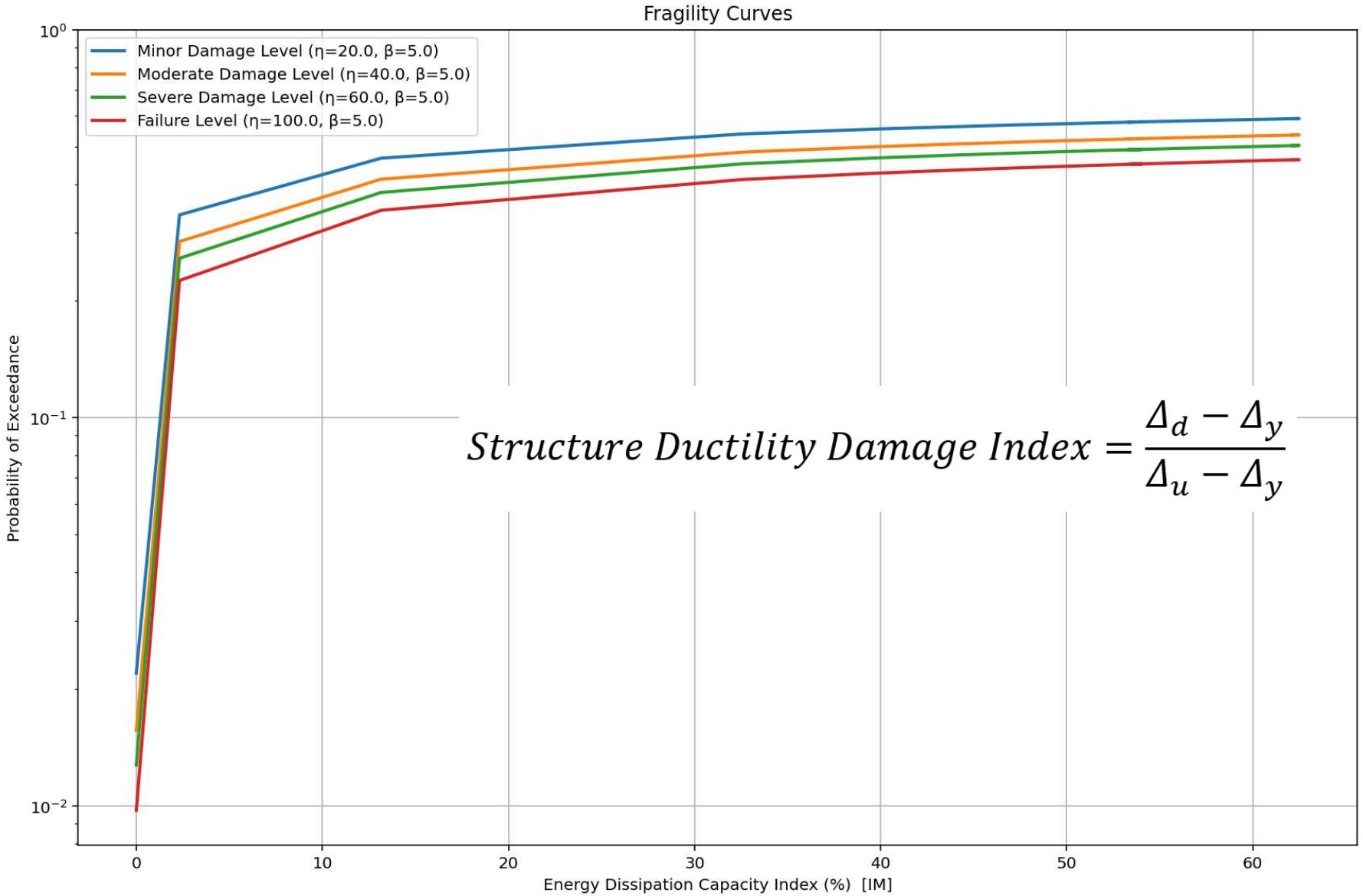


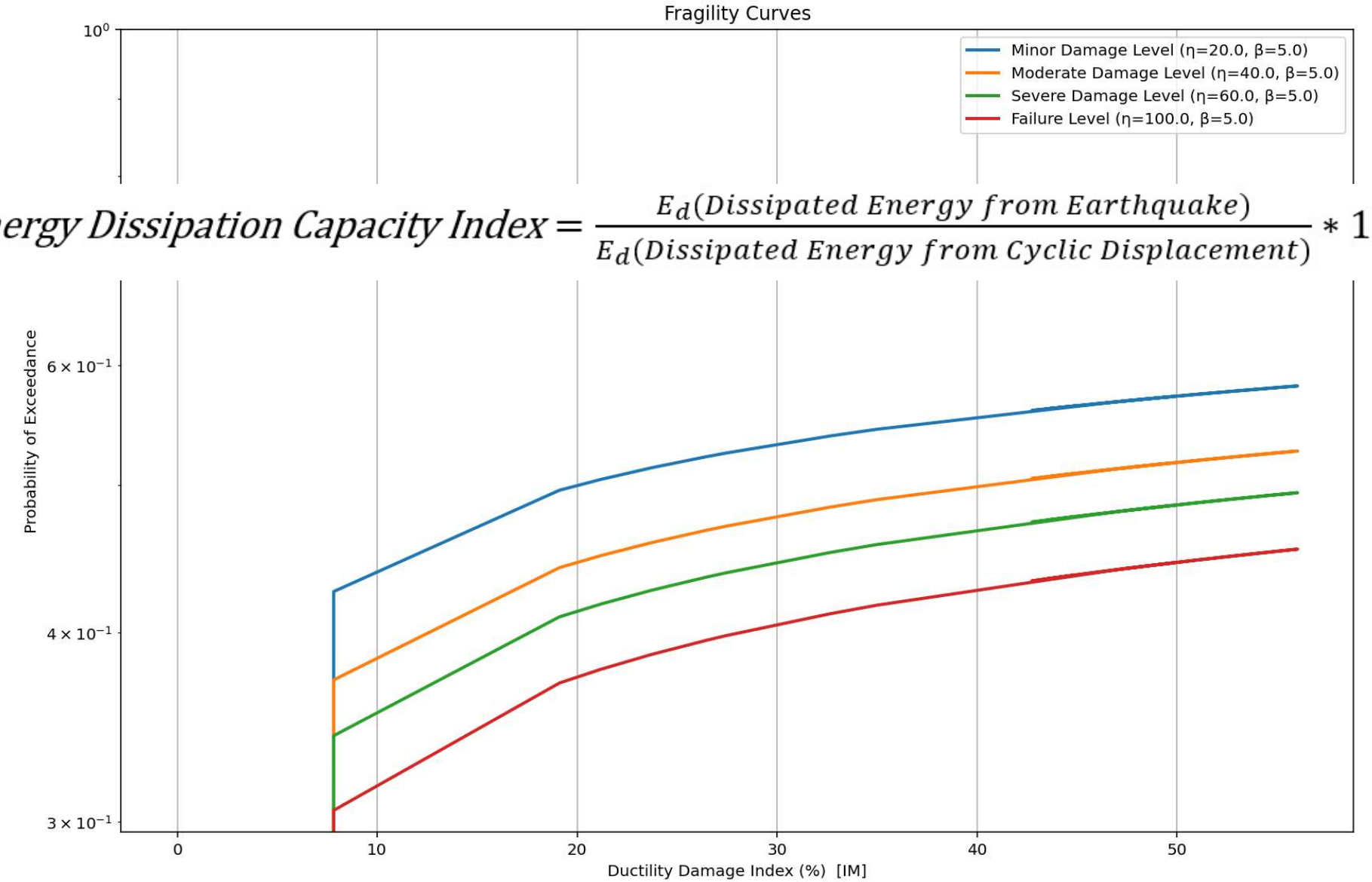
$$Energy\ Dissipation\ Capacity\ Index = \frac{E_d(Dissipated\ Energy\ from\ Earthquake)}{E_d(Dissipated\ Energy\ from\ Cyclic\ Displacement)} * 100$$


Equivalent viscous damping ratio Spectrum
Mean
Median
Q1 (25%)
Q3 (75%)
Mean ± Std
$\xi_{eq} = \frac{E_d}{4\pi E_s} * 100$
Equivalent viscous damping ratio [%]
Structural period [s]
0
50
100
150
200
250
300
350
0
1
2
3
4
5

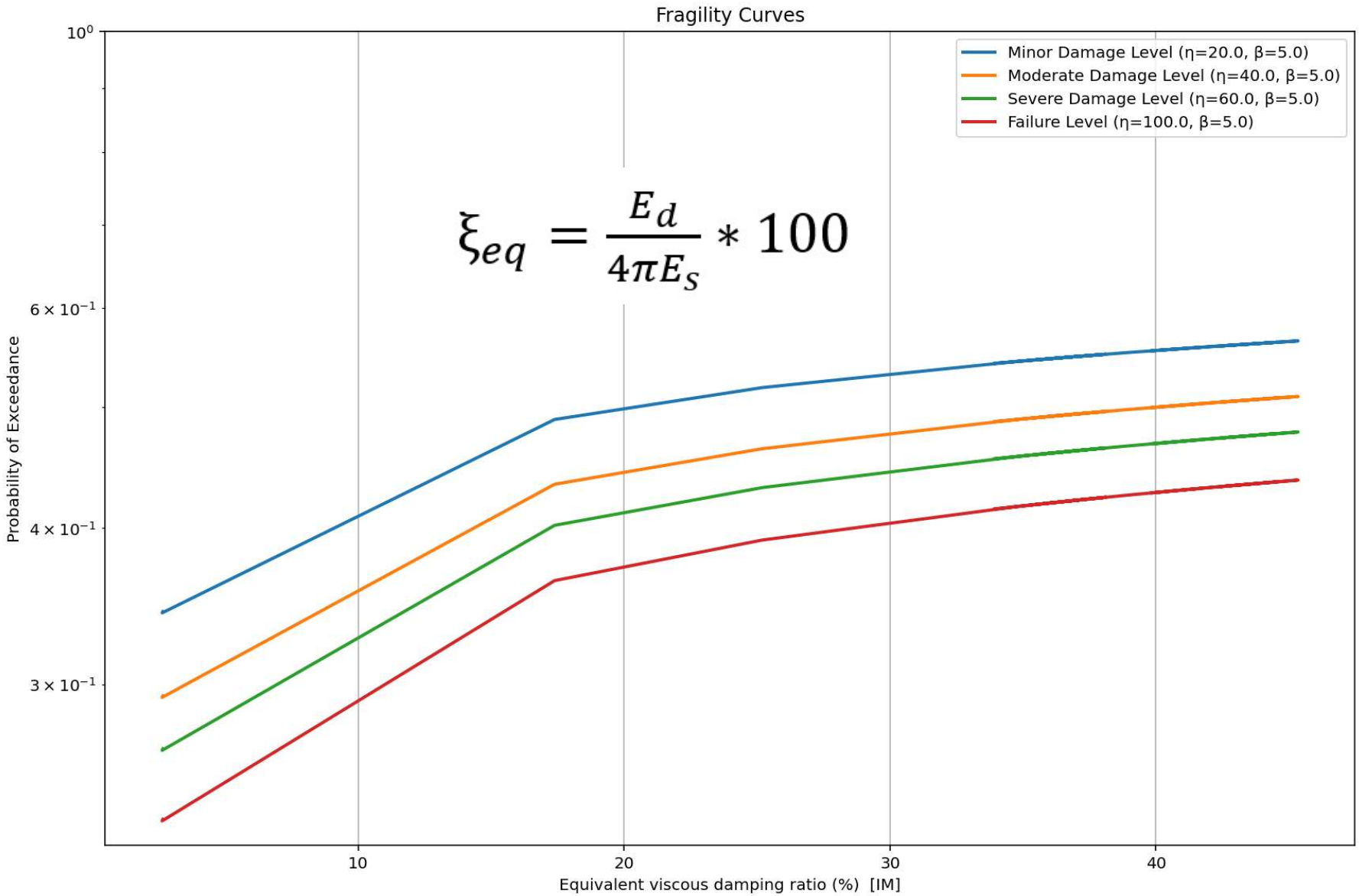

Last Analysis Structural Response + Ground Motion ::: MAX. ABS. : 16.3489
Acceleration (g)
15
10
5
0
−5
−10
−15
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

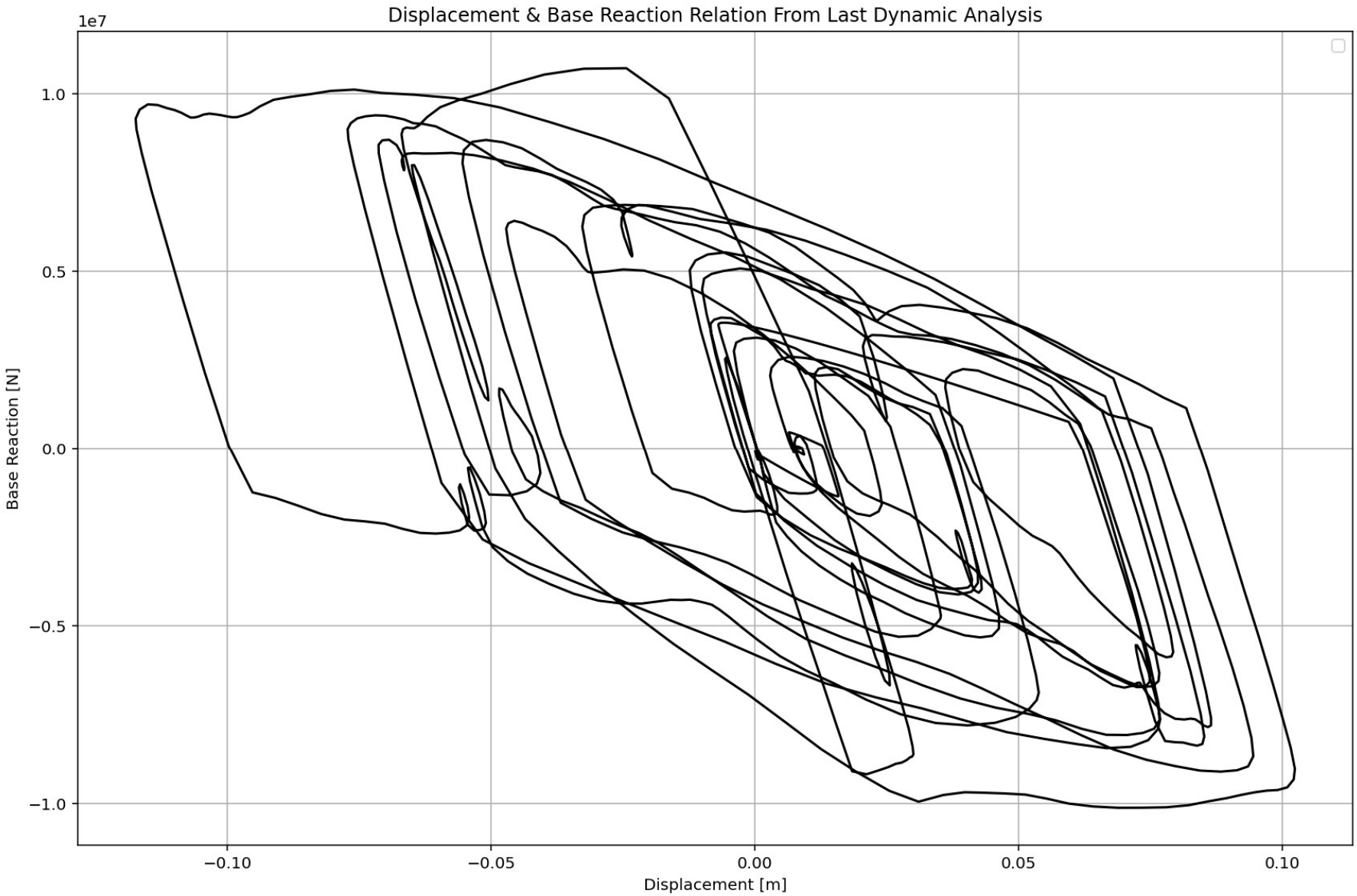

Fragility Curves
DS1_Slight (η=0.15, β=0.05)
DS2_Moderate (η=0.3, β=0.05)
DS3_Extensive (η=0.6, β=0.05)
DS4_Complete (η=1.0, β=0.05)
Probability of Exceedance
10^0
10^{-1}
15
20
25
30
35
40
45
50
Peak Ground Acceleration (g) [IM]

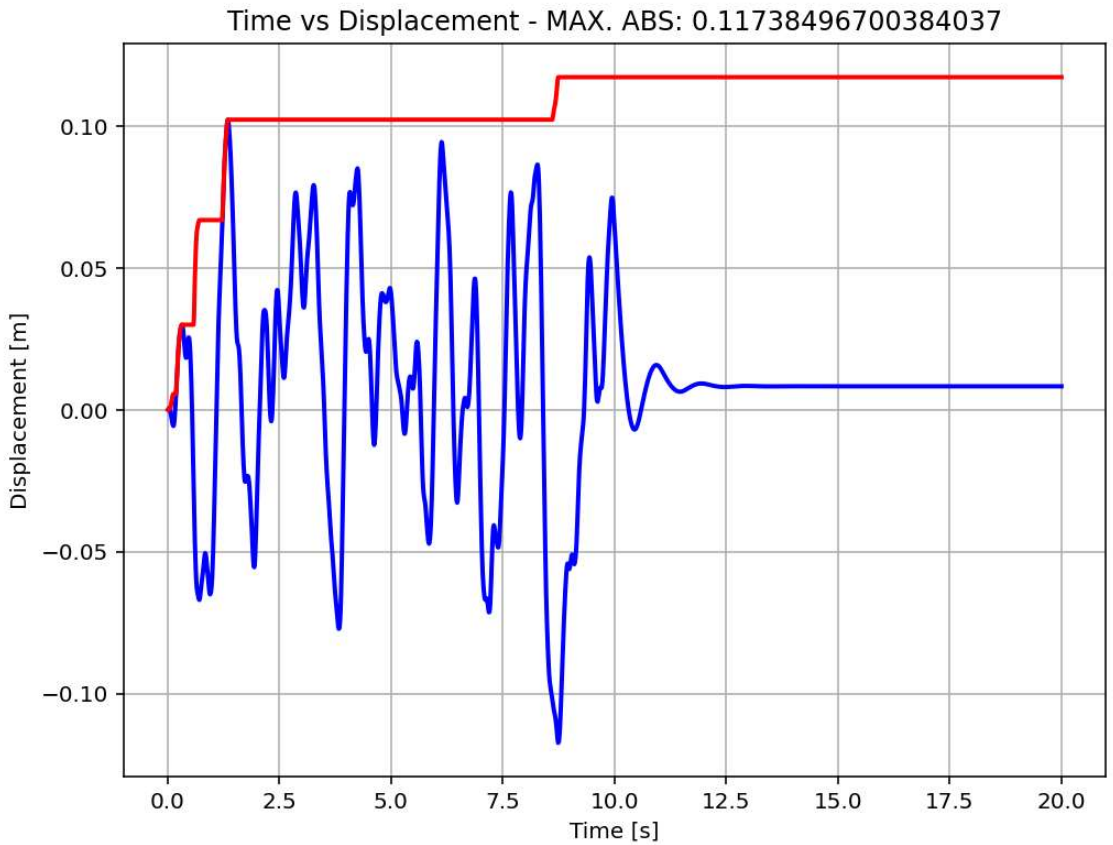

Fragility Curves
Minor Damage Level (η=20.0, β=5.0)
Moderate Damage Level (η=40.0, β=5.0)
Severe Damage Level (η=60.0, β=5.0)
Failure Level (η=100.0, β=5.0)
Probability of Exceedance
10^0
10^{-1}
10^{-2}
0
10
20
30
40
50
60
Energy Dissipation Capacity Index (%) [IM]
$Structure\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$

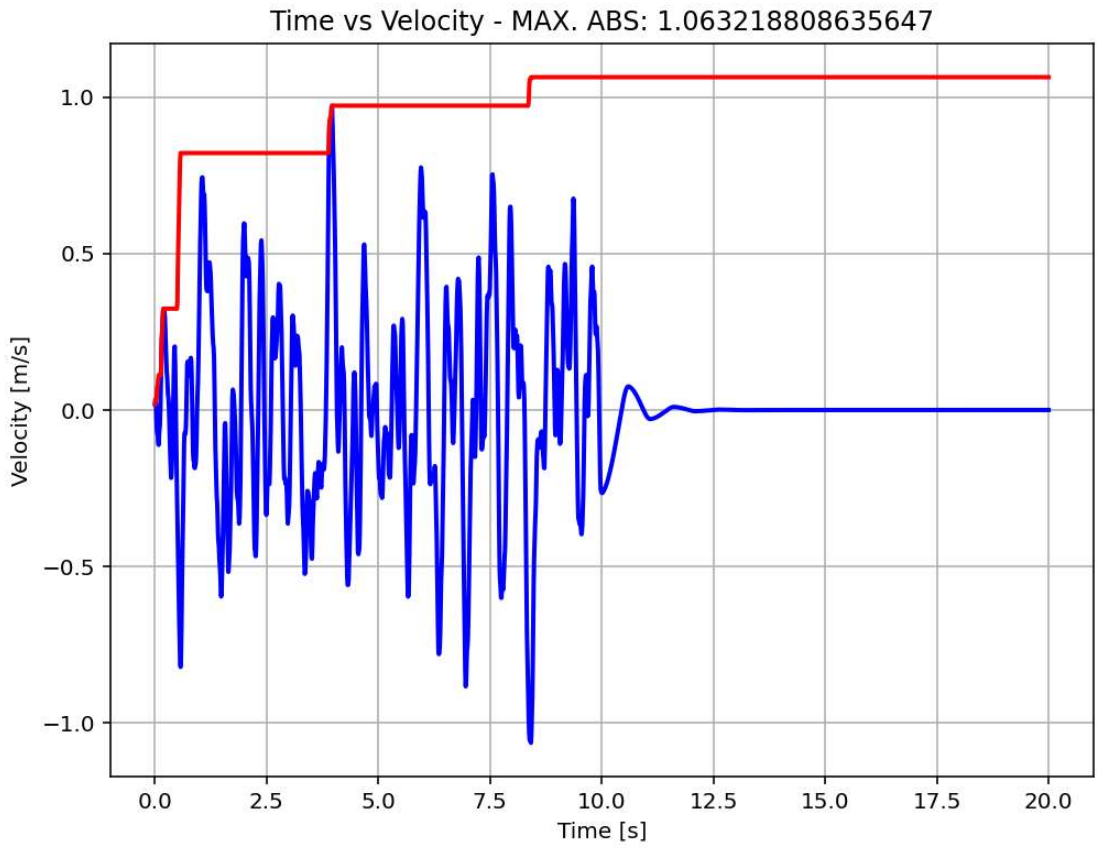


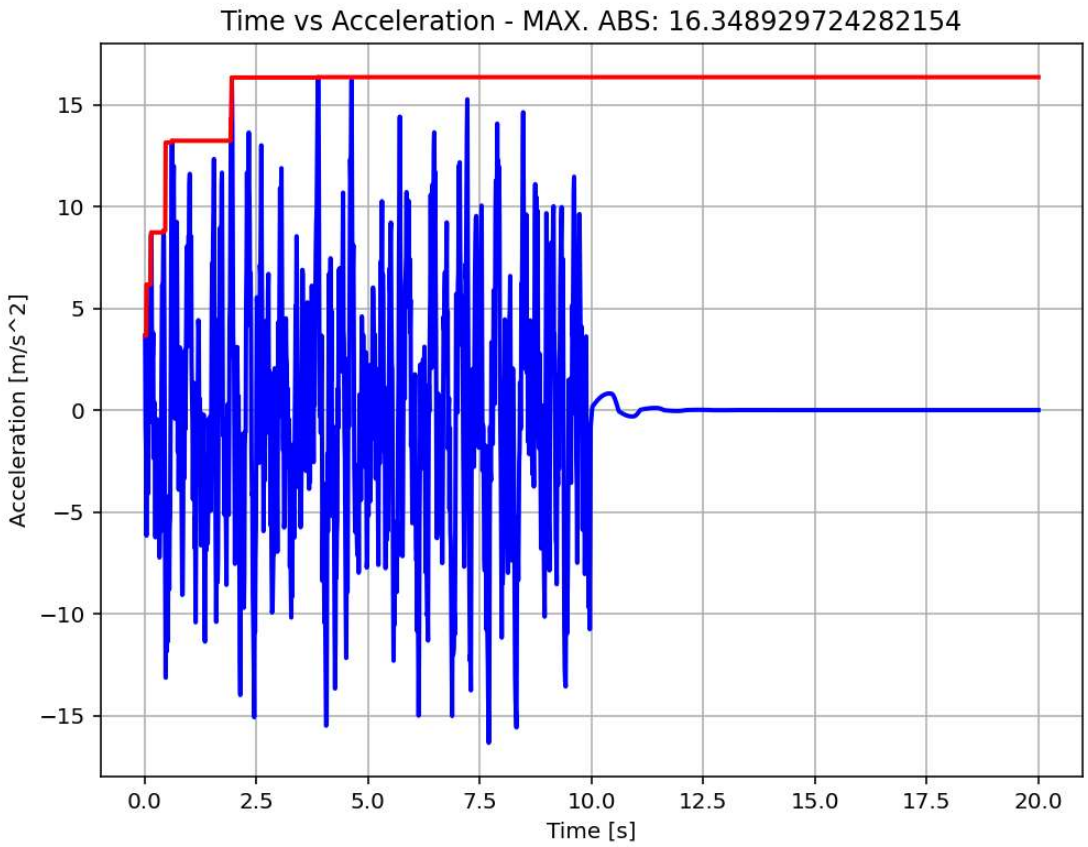
$$Energy\ Dissipation\ Capacity\ Index = \frac{E_d(Dissipated\ Energy\ from\ Earthquake)}{E_d(Dissipated\ Energy\ from\ Cyclic\ Displacement)} * 100$$

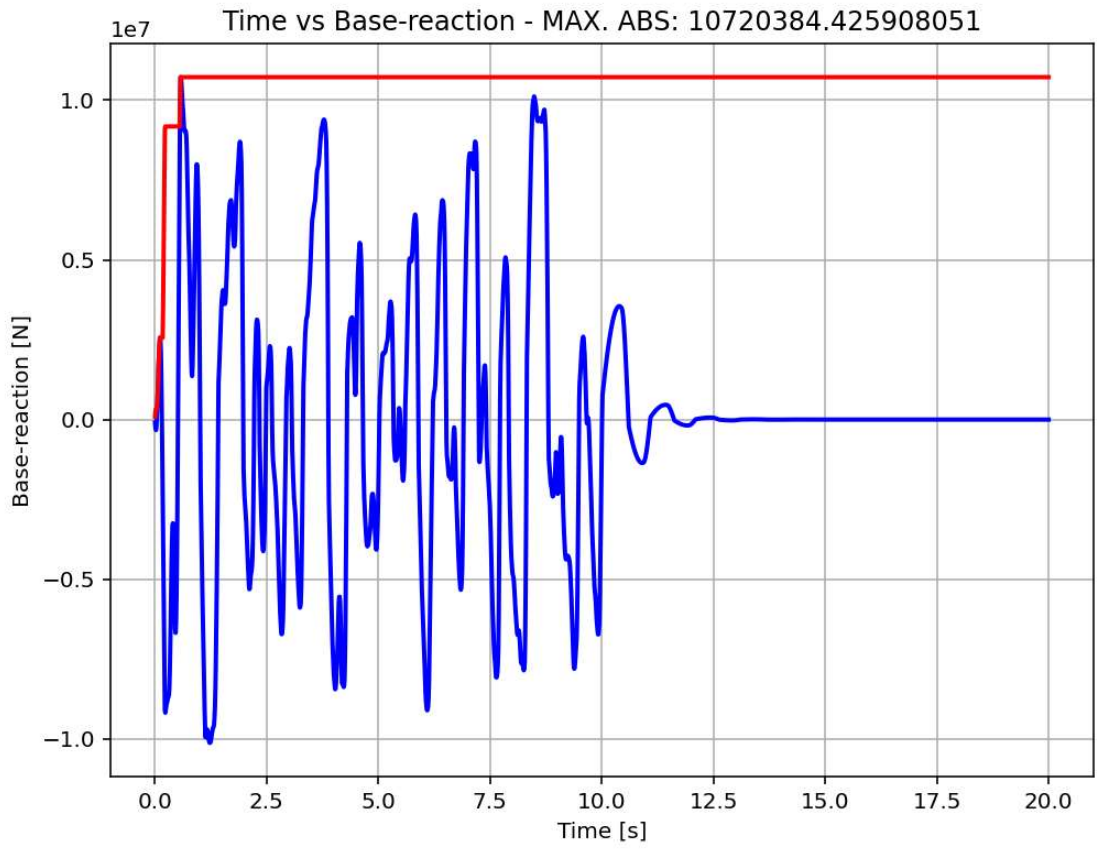

Fragility Curves
$\xi_{eq} = \frac{E_d}{4\pi E_s} * 100$
Minor Damage Level (η=20.0, β=5.0)
Moderate Damage Level (η=40.0, β=5.0)
Severe Damage Level (η=60.0, β=5.0)
Failure Level (η=100.0, β=5.0)
Probability of Exceedance
10^0
6×10^{-1}
4×10^{-1}
3×10^{-1}
10
20
30
40
Equivalent viscous damping ratio (%) [IM]

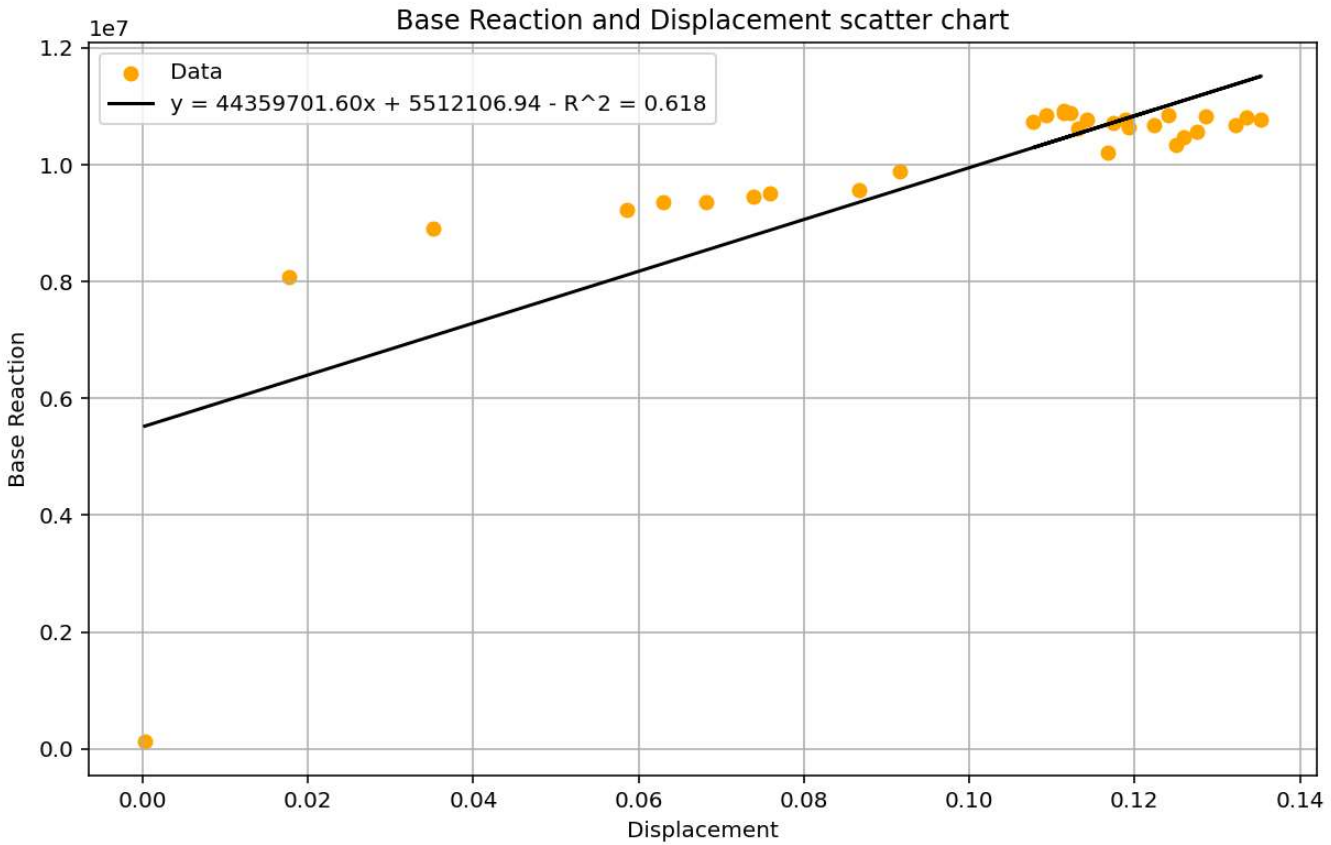

Displacement & Base Reaction Relation From Last Dynamic Analysis
1e7
1.0
0.5
0.0
−0.5
−1.0
Base Reaction [N]
−0.10
−0.05
0.00
0.05
0.10
Displacement [m]

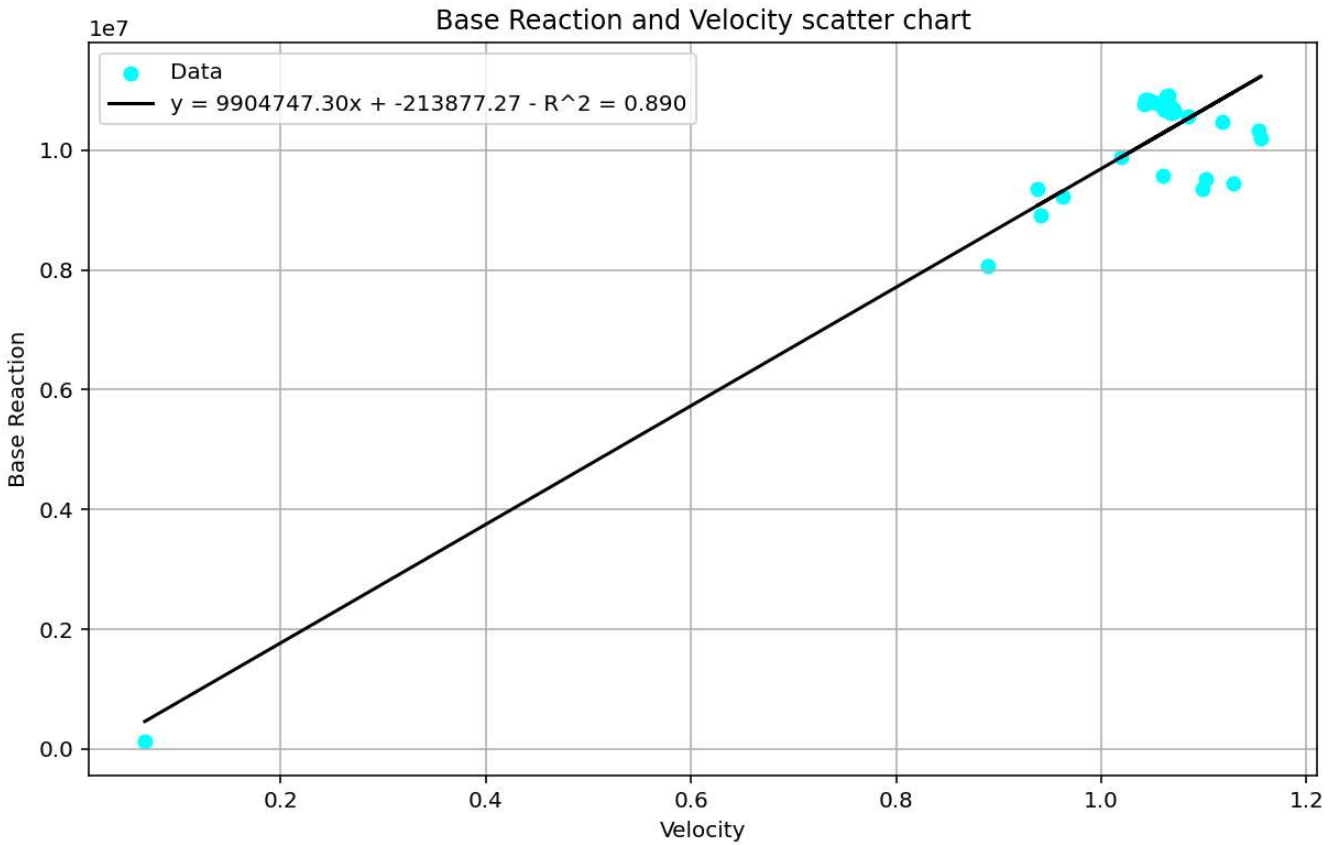

Time vs Displacement - MAX. ABS: 0.11738496700384037
Displacement [m]
0.10
0.05
0.00
−0.05
−0.10
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

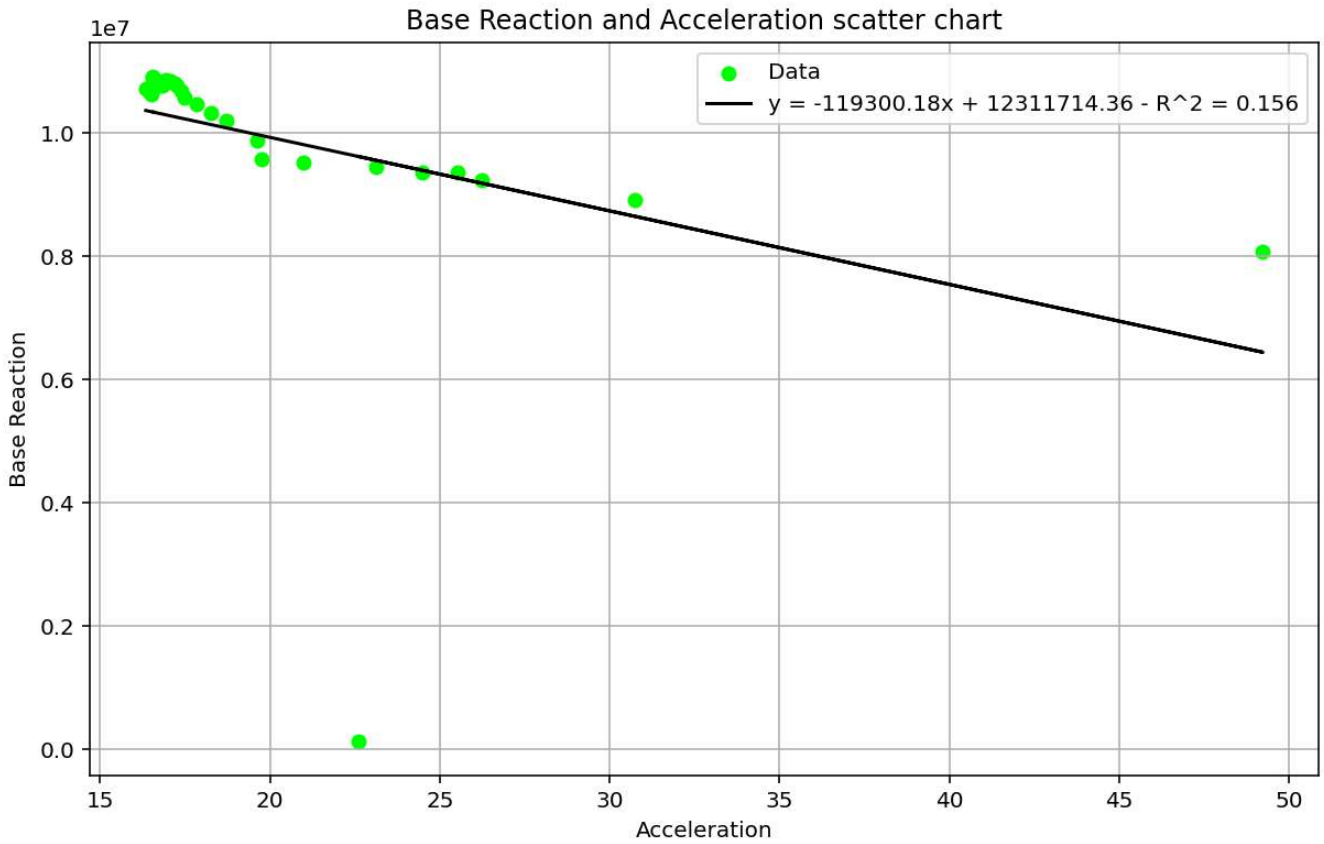

Time vs Velocity - MAX. ABS: 1.063218808635647
Velocity [m/s]
Time [s]
1.0
0.5
0.0
−0.5
−1.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

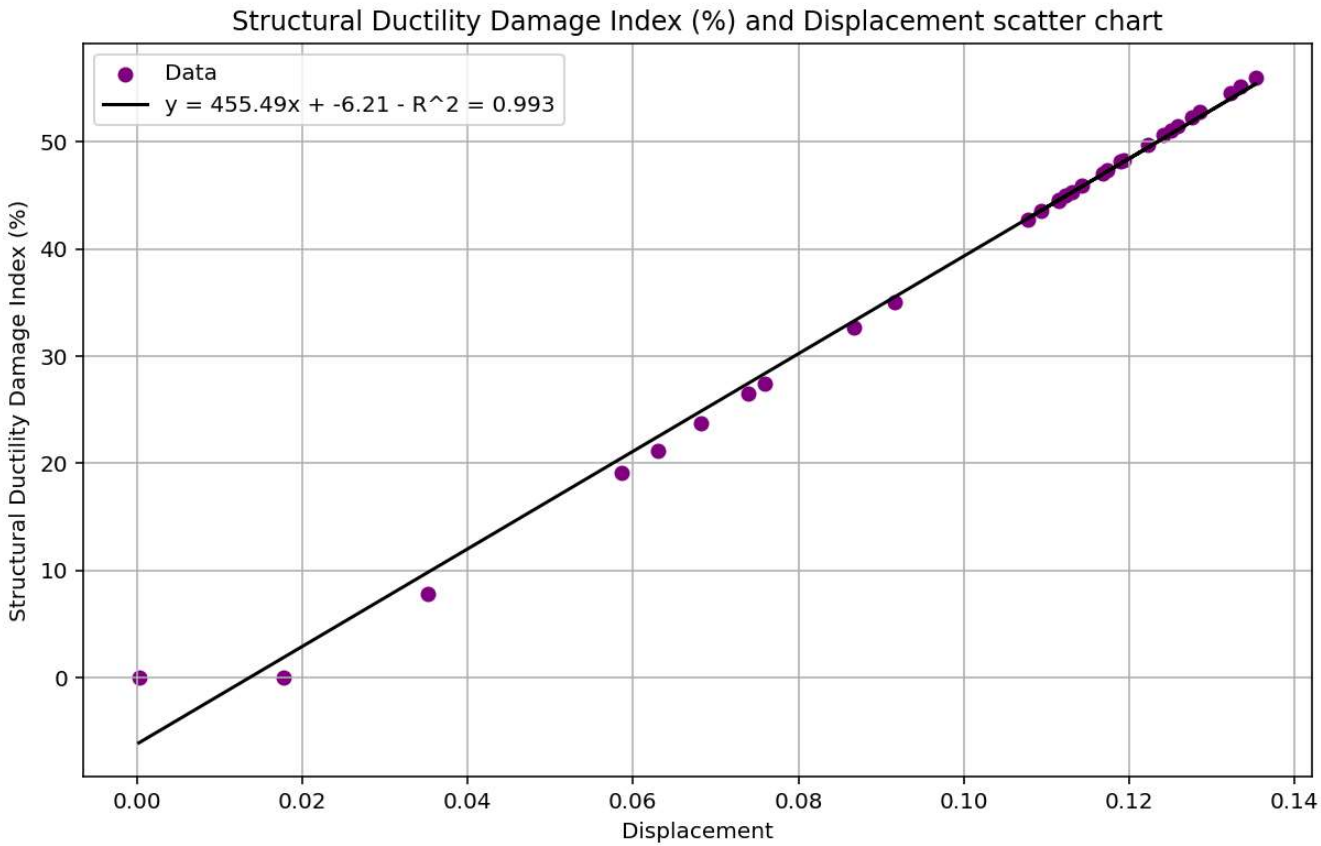

Time vs Acceleration - MAX. ABS: 16.348929724282154
Acceleration [m/s^2]
Time [s]
15
10
5
0
−5
−10
−15
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

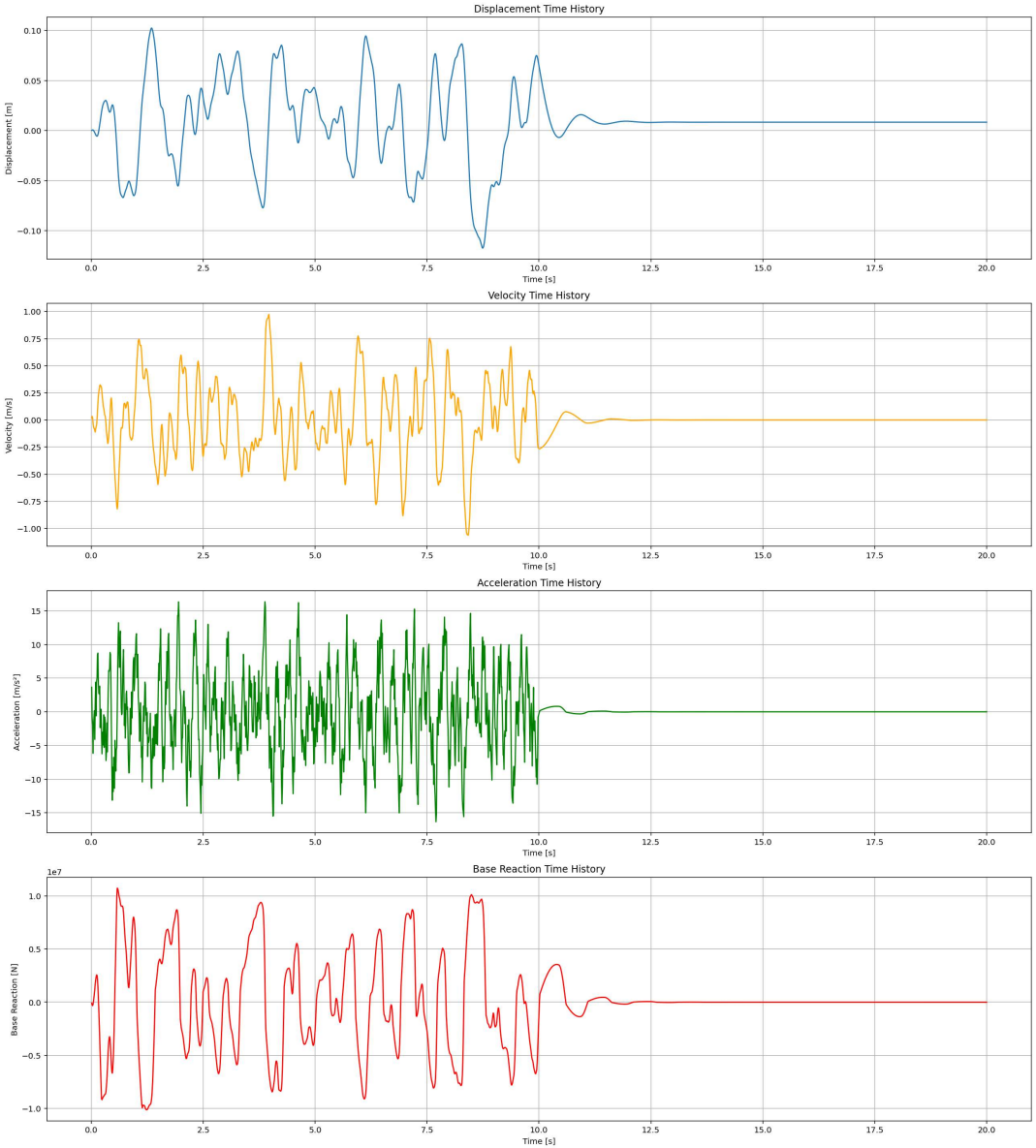

Time vs Base-reaction - MAX. ABS: 10720384.425908051
1e7
1.0
0.5
0.0
−0.5
−1.0
Base-reaction [N]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

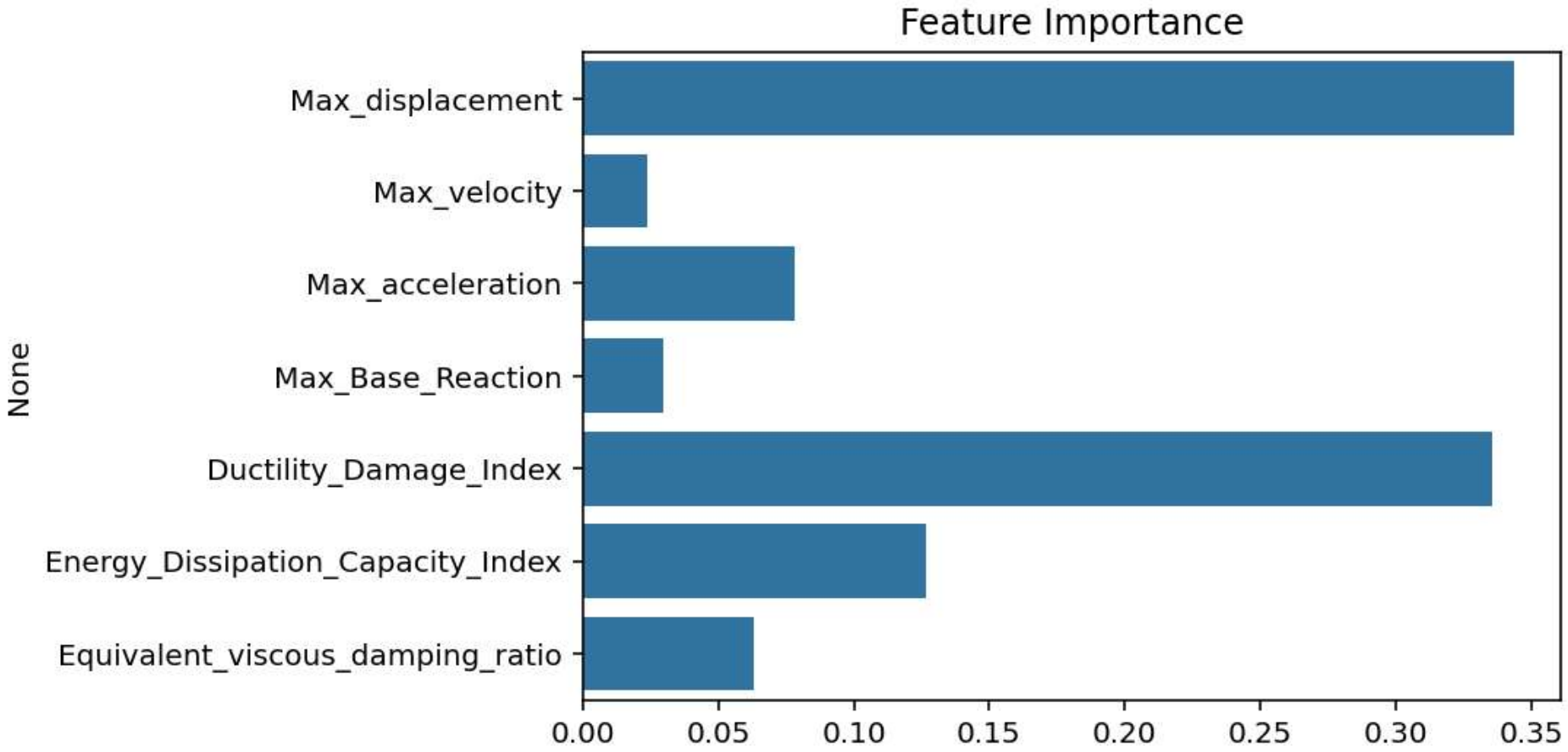

Base Reaction and Displacement scatter chart
1e7
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Base Reaction
Data
y = 44359701.60x + 5512106.94 - R^2 = 0.618
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
Displacement

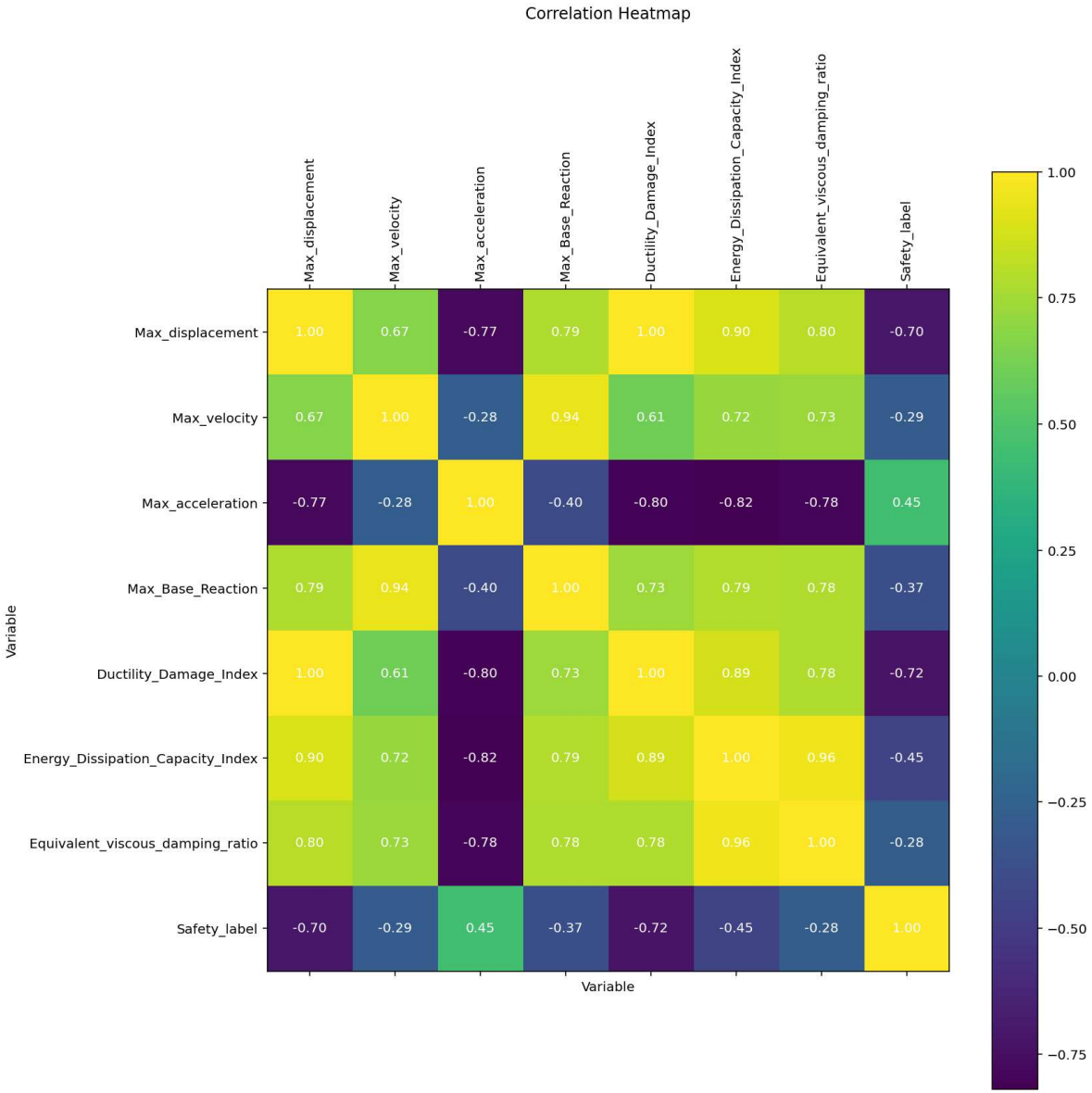

Base Reaction and Velocity scatter chart
1e7
Data
y = 9904747.30x + -213877.27 - R^2 = 0.890
Base Reaction
Velocity
0.0
0.2
0.4
0.6
0.8
1.0
0.2
0.4
0.6
0.8
1.0
1.2


Base Reaction and Acceleration scatter chart
1e7
Data
y = -119300.18x + 12311714.36 - R^2 = 0.156
Base Reaction
Acceleration
0.0
0.2
0.4
0.6
0.8
1.0
15
20
25
30
35
40
45
50


Structural Ductility Damage Index (%) and Displacement scatter chart
Data
y = 455.49x + -6.21 - R^2 = 0.993
Structural Ductility Damage Index (%)
Displacement
0
10
20
30
40
50
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14


Displacement Time History
Displacement [m]
Time [s]
Velocity Time History
Velocity [m/s]
Time [s]
Acceleration Time History
Acceleration [m/s²]
Time [s]
Base Reaction Time History
1e7
Base Reaction [N]
Time [s]


Feature Importance
Max_displacement
Max_velocity
Max_acceleration
Max_Base_Reaction
Ductility_Damage_Index
Energy_Dissipation_Capacity_Index
Equivalent_viscous_damping_ratio
None
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35


Correlation Heatmap
Max_displacement
Max_velocity
Max_acceleration
Max_Base_Reaction
Ductility_Damage_Index
Energy_Dissipation_Capacity_Index
Equivalent_viscous_damping_ratio
Safety_label
Variable
Variable
1.00 0.67 -0.77 0.79 1.00 0.90 0.80 -0.70
0.67 1.00 -0.28 0.94 0.61 0.72 0.73 -0.29
-0.77 -0.28 1.00 -0.40 -0.80 -0.82 -0.78 0.45
0.79 0.94 -0.40 1.00 0.73 0.79 0.78 -0.37
1.00 0.61 -0.80 0.73 1.00 0.89 0.78 -0.72
0.90 0.72 -0.82 0.79 0.89 1.00 0.96 -0.45
0.80 0.73 -0.78 0.78 0.78 0.96 1.00 -0.28
-0.70 -0.29 0.45 -0.37 -0.72 -0.45 -0.28 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75